



Intelligent Systems

Final Exam Crash Course | Mara Spadon | 2023

Welcome!



Mara Spadon

About me

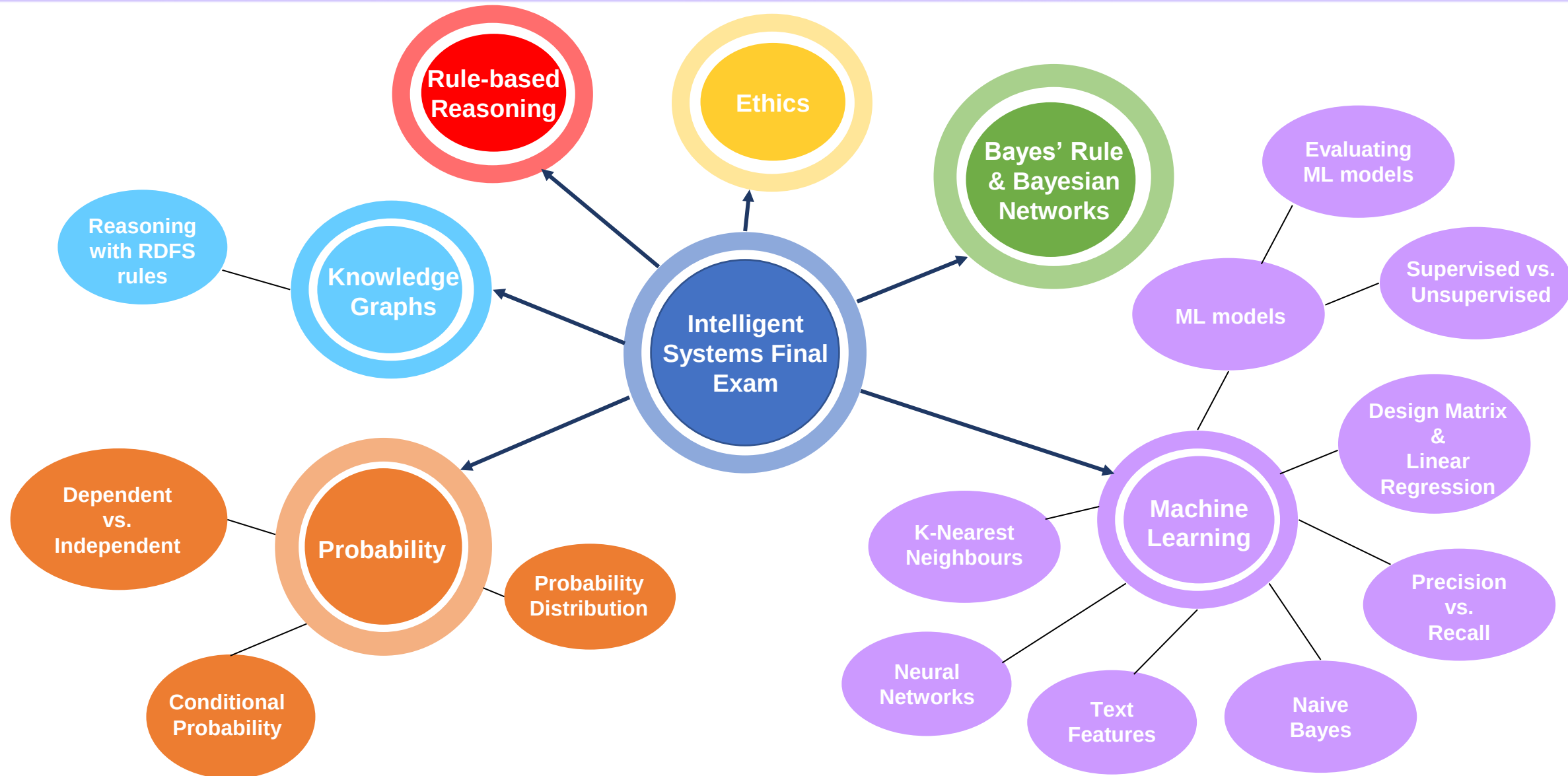
- Bachelor's Degree AI
- TA Intelligent Systems & Athena Studies
- Past experience with teaching this course

WhatsApp

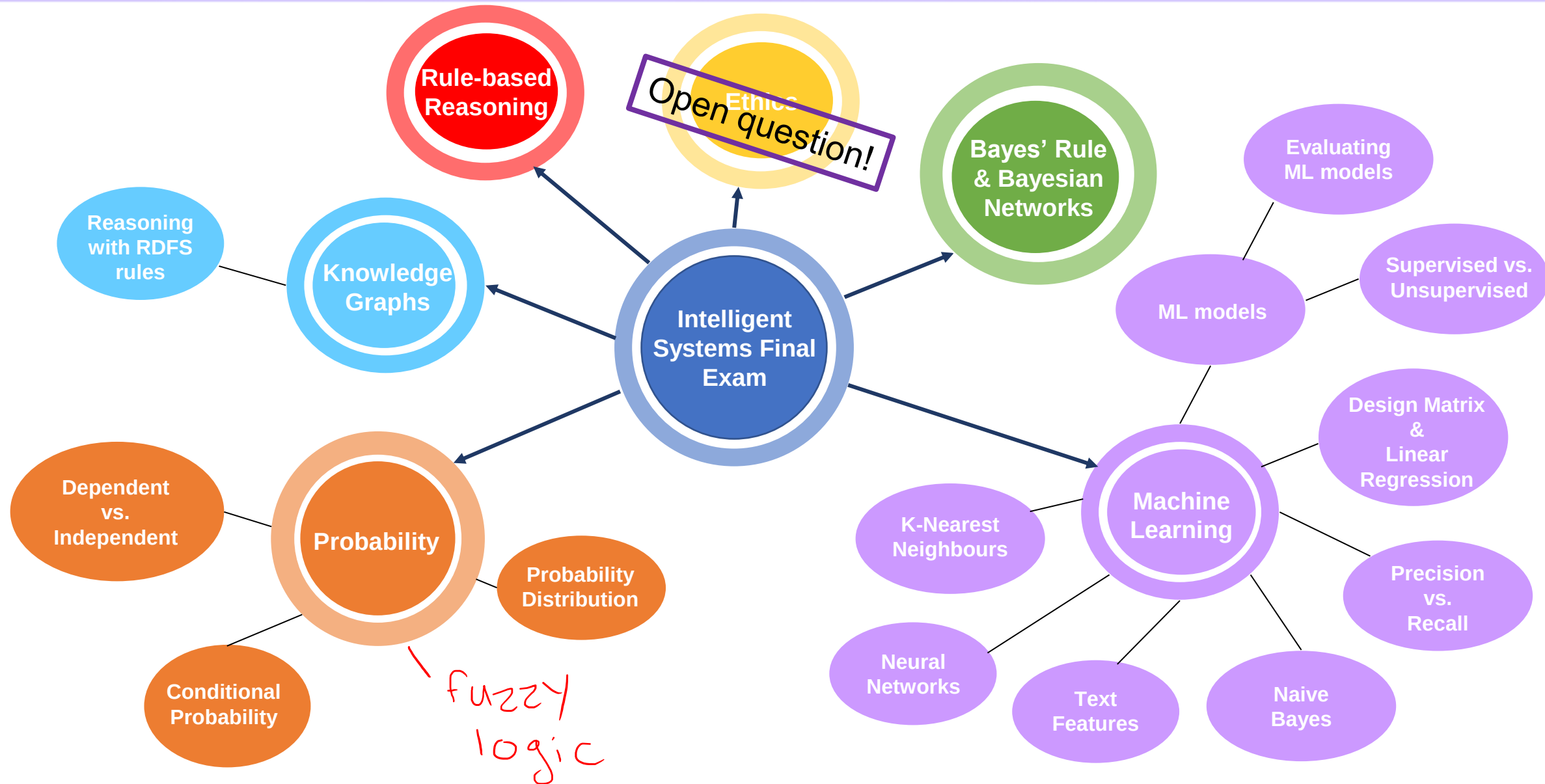


Slides will
be sent
in the
group! ▽

Exam analysis



Exam analysis



Pro-active participation

- Camera's on (optional)
- Raise your hand to ask questions / Ask questions in the chat!
- Be interactive!
- Take notes!
- It is okay to make mistakes
- Slides will be sent afterwards



Structure



Structure based on past exams



Learn the most by asking questions



Active learning with old exam questions is the basis of our trainings

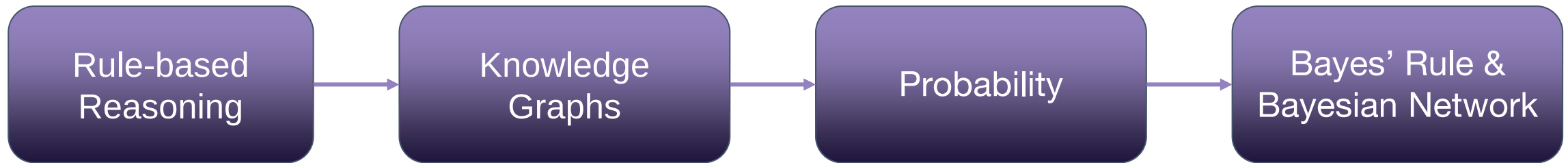


24/7 WhatsApp aid

Full Planning

Sunday	○ Rule-based Reasoning ○ Knowledge Graphs ○ Probability ○ Bayes' Rule & Bayesian Networks
Monday	Machine Learning: ○ ML models and evaluations ○ Text Features & K-NN ○ Design Matrix, Linear Regression & Neural Networks ○ Final practice: Exam Questions

Planning Day 1



Rule-based Reasoning

Rule-based Reasoning

Rule-based reasoning: different way of proving entailment

Remember DPLL?

Rule-based Reasoning: Inference Rules

An **inference rule** is a logical form consisting of a function which takes **premises**, analyzes their syntax, and returns a **conclusion** (or conclusions).

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta, \alpha}$$

Rule-based Reasoning: Inference Rules

An **inference rule** is a logical form consisting of a function which takes **premises**, analyzes their syntax, and returns a **conclusion** (or conclusions).

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

This is NOT the same as:
 $((A \rightarrow B) \wedge A) \rightarrow B$

Rule-based Reasoning: Inference Rules

Modens Ponens
$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta} \text{ or: } \alpha$$

And-elimination: from a conjunction any conjunct can be inferred:

$$\frac{\alpha \wedge \beta}{\alpha \text{ or: } \beta}$$

Logical equivalences can be used as rules:

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

$(\dots) \wedge (\dots)$

Do these rules look familiar?

Rule-based Reasoning: Inference Rules

$$\begin{aligned}(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{de Morgan} \\\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{de Morgan} \\(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge\end{aligned}$$

Rule-based Reasoning

Rule-based reasoning: different way of proving entailment

How do we actually prove the entailment? $KB \models \alpha$?

Rule-based Reasoning

Rule-based reasoning: different way of proving entailment

How do we actually prove the entailment? $KB \models \alpha$?

Forward Chaining!

/checking

Forward Chaining: Step Plan

$KB \models \alpha$?

1. Write down all the facts that can be derived from the KB by applying all the inference rules exhaustively

→ This set of facts is called the **closure** (KB^*)

$$KB^* = \{ \dots \}$$

2. If the KB^* includes α , then α is entailed by the knowledge base
↳ closure

When $\alpha \in KB^*$, we show that $KB \vdash_f \alpha$, which implies $KB \models \alpha$

Rule-based Reasoning: Recap Question 1

Prove or disprove whether
 $KB = \{C \rightarrow \neg(A \vee \neg B), A\}$ entails $\neg C$,
 using Forward Chaining/ Checking.

Give the rules that you apply, and write
 down the closure.

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$
 $(\alpha \vee \beta) \equiv (\beta \vee \alpha)$ commutativity of $\vee$
 $((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$ associativity of $\wedge$
 $((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$ associativity of $\vee$
 $\neg(\neg\alpha) \equiv \alpha$ double-negation elimination
 $(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$ contraposition
 $(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$ implication elimination
 $(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$ biconditional elimination
 $\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$ de Morgan
 $\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$ de Morgan
 $(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$ distributivity of $\wedge$ over $\vee$
 $(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$ distributivity of $\vee$ over $\wedge$

$KB = \{C \rightarrow \neg(A \vee \neg B), A\}$, $KB \models \neg C$?

$\neg C \vee \neg(A \vee \neg B)$ impl elimination

$\neg C \vee (\neg A \wedge B)$ de Morgan

$(\neg C \vee \neg A) \wedge (\neg C \vee B)$ distr.

$\neg C \vee \neg A$, $\neg C \vee B$ and elimination

$KB^* = \{A, C \rightarrow \neg(A \vee \neg B), \neg C \vee \neg(A \vee \neg B), \neg C \vee (\neg A \wedge B), \dots$

$\neg\neg B = B$

$$(\neg C \vee \neg A) \wedge (\neg C \vee B)$$

$\neg C$ is not in our $\mathcal{KB}^*$

so $\neg C$ is not entailed by
our $\mathcal{KB}$

$$\mathcal{KB}^* = \{ \dots, (\neg C \vee \neg A) \wedge (\neg C \vee B), \neg C \vee \neg A, \neg C \vee B \}$$

~~$\neg C$~~

Rule-based Reasoning: Recap Question 2

Prove or disprove whether
 $KB = \{C \rightarrow \neg(A \vee \neg B), C\}$ entails $\neg A$,
 using Forward Chaining/ Checking.

Give the rules that you apply, and write
 down the closure.

Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

And-elimination: from a conjunction any
 conjunct can be inferred:

$$\frac{\alpha \wedge \beta}{\alpha}$$

$$\frac{\alpha \wedge \beta}{\beta}$$

Logical equivalences can be used as rules:

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

$$KB = \{C \rightarrow \neg(A \vee \neg B), C\}$$

$$C, \neg(A \vee \neg B)$$

$$\neg A \wedge B$$

$$\downarrow$$

$$\neg A, B$$

entails $\neg A$

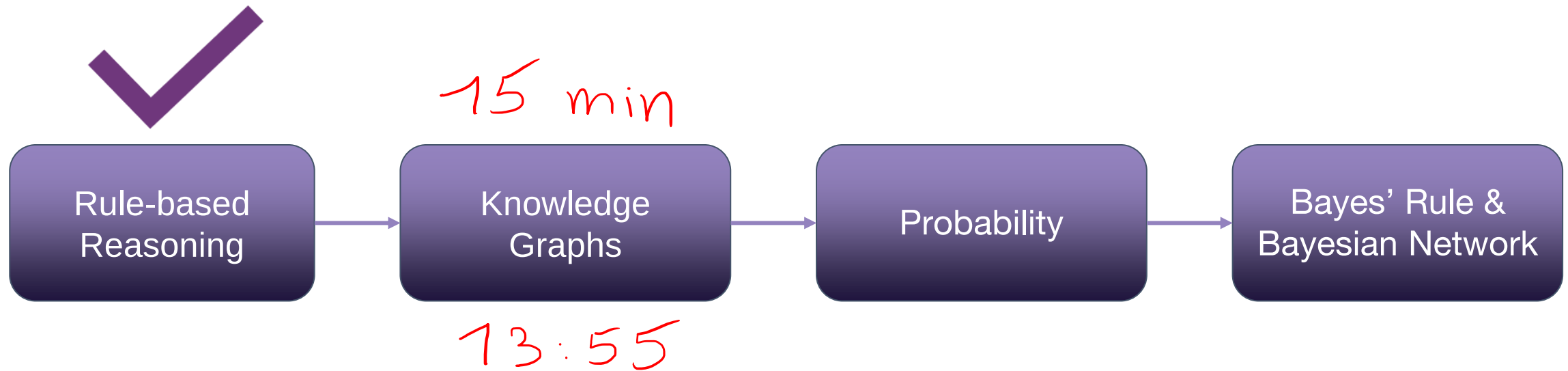
Modus Ponens
 de Morgan

and elimination

$$KB^* = \{C, C \rightarrow \neg(A \vee \neg B), \neg(A \vee \neg B), \neg A \wedge B, \neg A, B\}$$

$\neg A \in KB^*$? yes!
 so
 $KB \models \neg A$

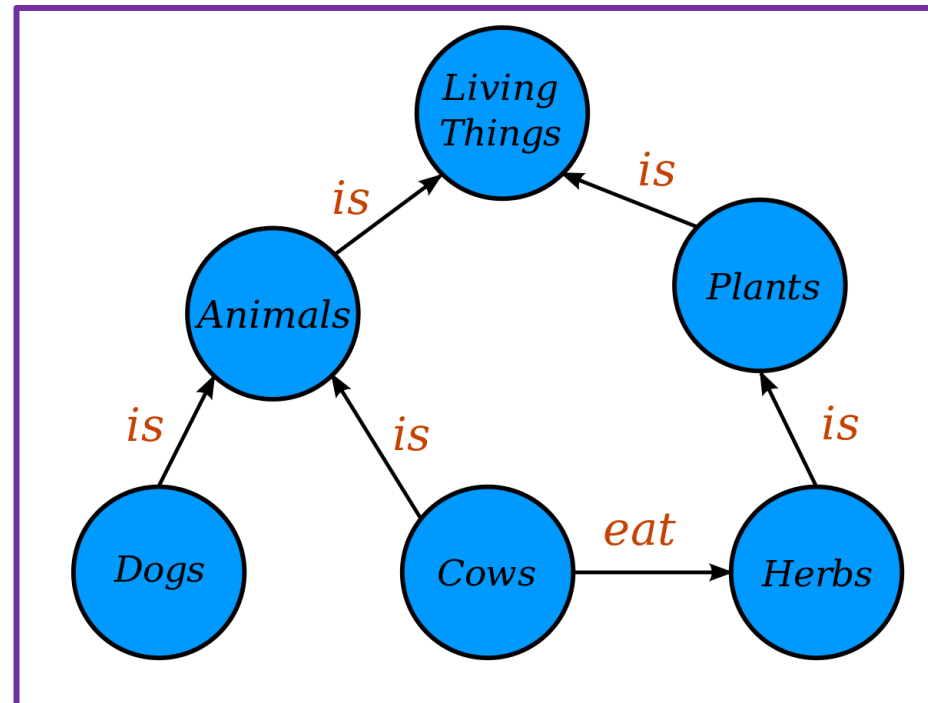
Planning Day 1



Knowledge Graphs

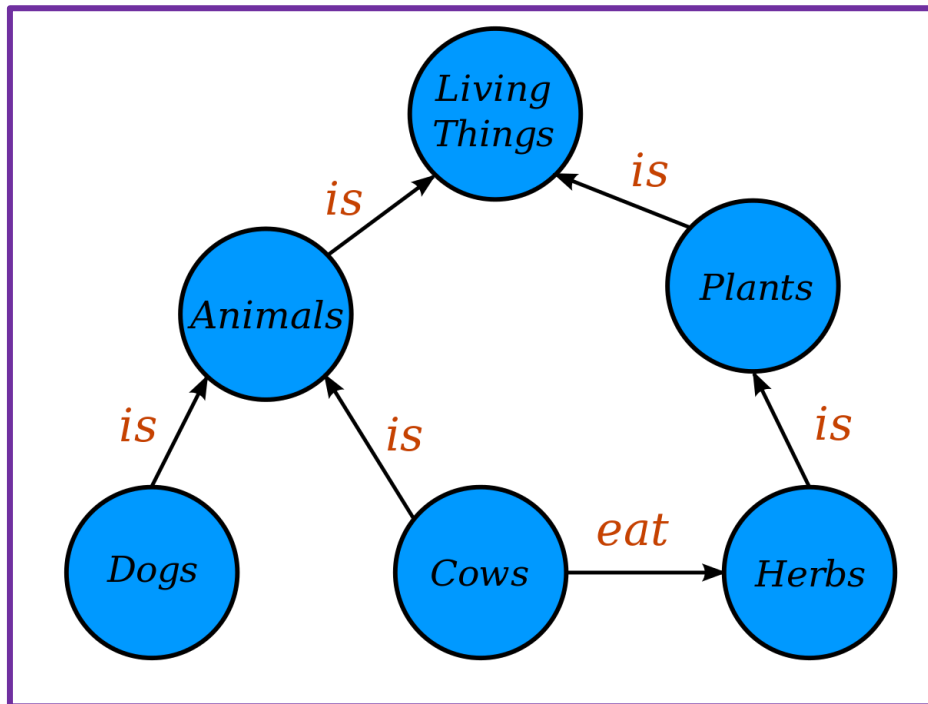
Knowledge Graphs

Knowledge graph: A digital structure that represents knowledge as concepts and the relationships between them.



Knowledge Graphs: Triples

Knowledge graph: A digital structure that represents knowledge as concepts and the relationships between them.

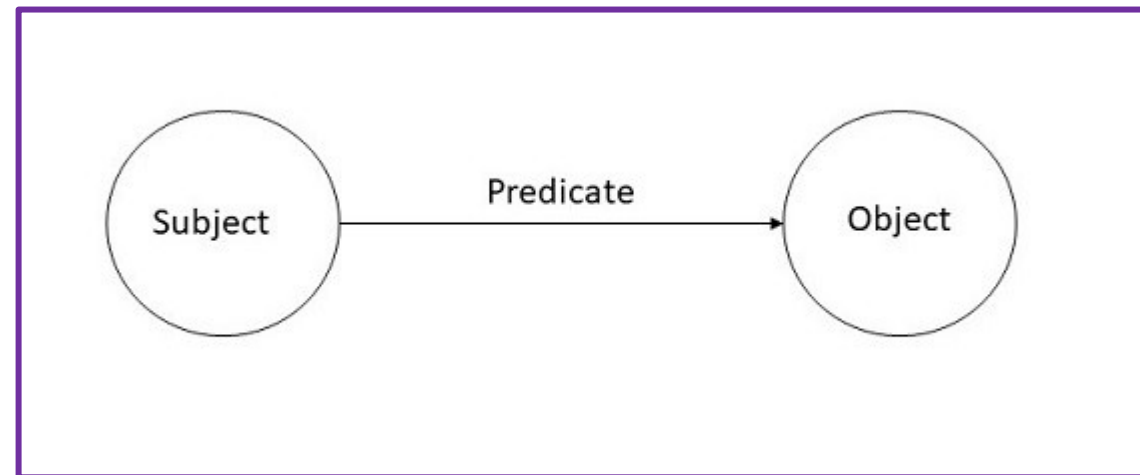


We can express the graph data using **triples** (facts):

{Animals is LivingThings, Plants is LivingThings, Dogs is Animals, Cows is Animals, Cows eat Herbs, Herbs is Plants, Plants is LivingThings}

Knowledge Graphs: Triples

We can express the knowledge graph data using **triples** (facts):



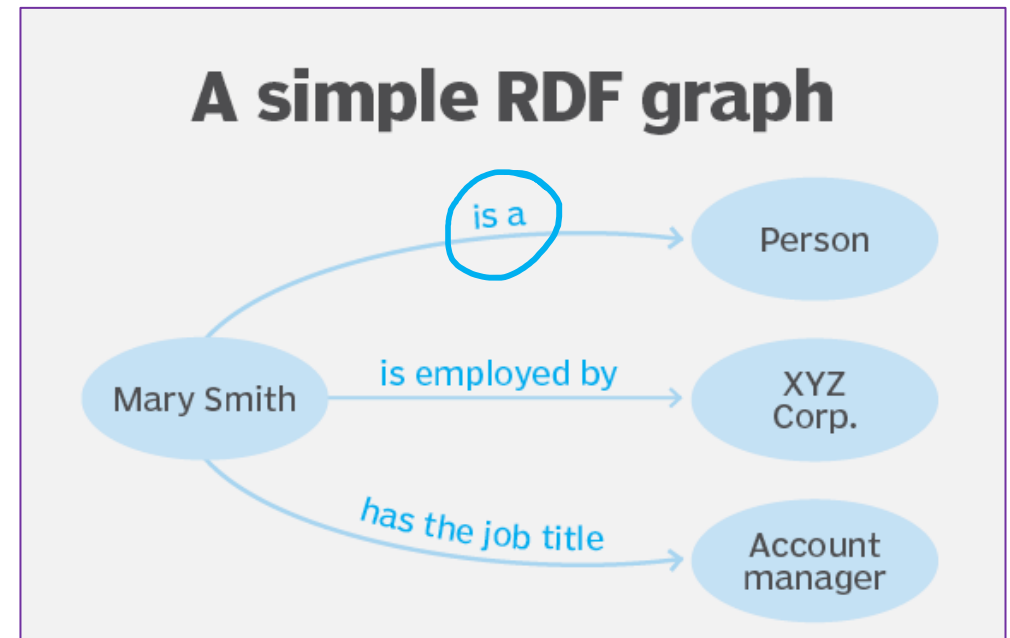
1. **Subject** is a resource being described by the triple.
2. **Predicate** describes the relationship between the subject and the object.
3. **Object** is a resource that is related to the subject.

Resource Description Framework (RDF) is a general framework for representing interconnected data on the web.

RDF statements express relationships between resources.

→ 'is a'

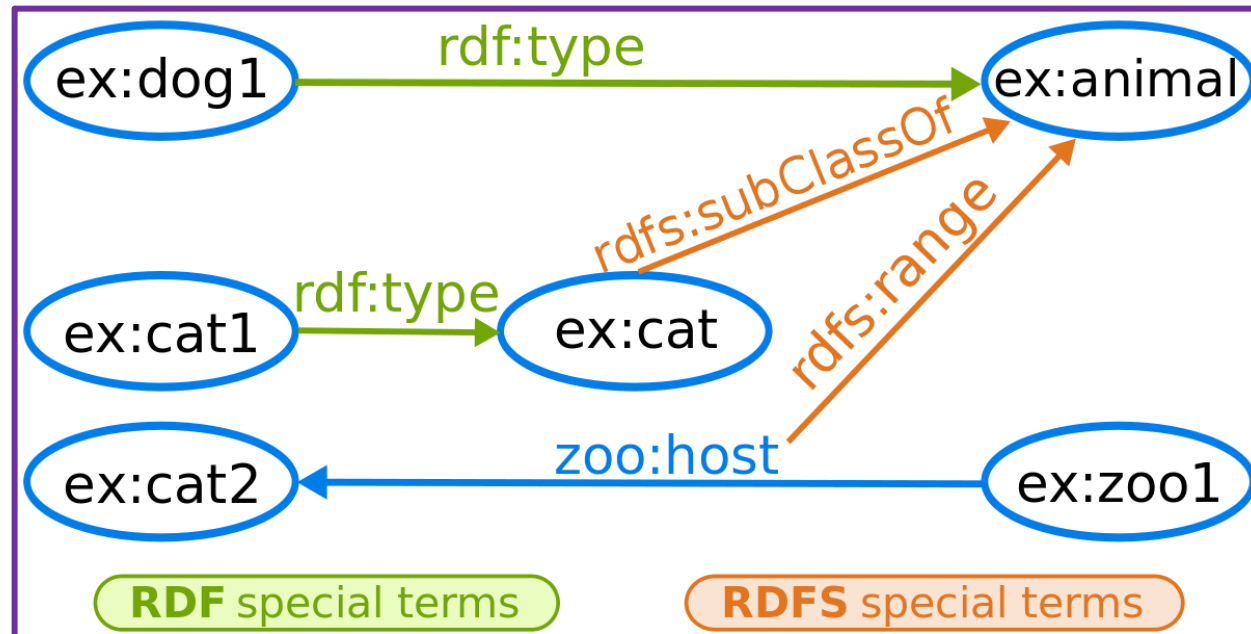
ex: Student rdf:type ex:Person
 subj. pred. obj.



Knowledge Graphs: RDFS

RDFS provides a mechanism for describing groups of related resources (RDF), and the relation between them.

Example of these properties are *classes* or *sub-classes*



Knowledge Graphs: Schema Vocabulary

Specify that something is a **member** of a **class**

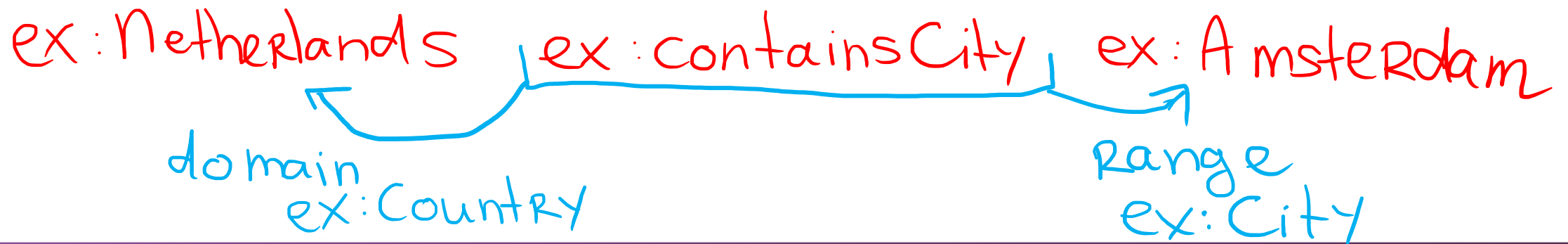
`ex:Netherlands` `rdf:type` `ex:EuropeanCountry`

Specify that something is a **subclass** of another **class**

`ex:EuropeanCountry` `rdfs:subClassOf` `ex:Country`

Specify that some **property** always relates members of **specific classes**

`ex:containsCity` `rdfs:domain` `ex:Country`
`ex:containsCity` `rdfs:range` `ex:City`



Knowledge Graphs: Entailment Rules

1. $s \quad p \quad o \implies \begin{matrix} s & \text{rdf:type} & \text{rdfs:Resource} \\ p & \text{rdf:type} & \text{rdf:Property} \\ o & \text{rdf:type} & \text{rdfs:Resource} \\ \text{OR} \\ o & \text{rdf:type} & \text{rdfs:Literal} \end{matrix}$

2. $s \text{ rdf:type } o \implies o \text{ rdf:type rdfs:Class}$

3. $s \text{ rdf:type rdfs:Class} \implies s \text{ rdfs:subClassOf } s$

4. $p \text{ rdf:type rdf:Property} \implies p \text{ rdfs:subPropertyOf } p$

$$5. \begin{matrix} s & \text{rdfs:subClassOf} & y \\ y & \text{rdfs:subClassOf} & z \end{matrix} \implies s \text{ rdfs:subClassOf } z$$

$$6. \begin{matrix} p & \text{rdfs:subPropertyOf} & q \\ q & \text{rdfs:subPropertyOf} & r \end{matrix} \implies p \text{ rdfs:subPropertyOf } r$$

$$7. \begin{matrix} s & p & o \\ p & \text{rdfs:subPropertyOf} & q \end{matrix} \implies s \quad q \quad o$$

$$8. \begin{matrix} s & p & o \\ p & \text{rdfs:domain} & x \end{matrix} \implies s \text{ rdf:type } x$$

$$9. \begin{matrix} s & p & o \\ p & \text{rdfs:range} & x \end{matrix} \implies o \text{ rdf:type } x$$

$$10. \begin{matrix} s & \text{rdf:type} & o \\ o & \text{rdfs:subClassOf} & t \end{matrix} \implies s \text{ rdf:type } t$$

$ex:containsCity \text{ rdfs:domain } ex:country$
 $ex:contains \text{ rdfs:range } ex:City$
 $ex:Netherlands \text{ } ex:containsCity \text{ } ex:Amsterdam$

$\implies ex:Netherlands \text{ rdf:type } ex:Country, ex:Ams \text{ rdf:type } ex:City$

Reasoning with RDFS rules

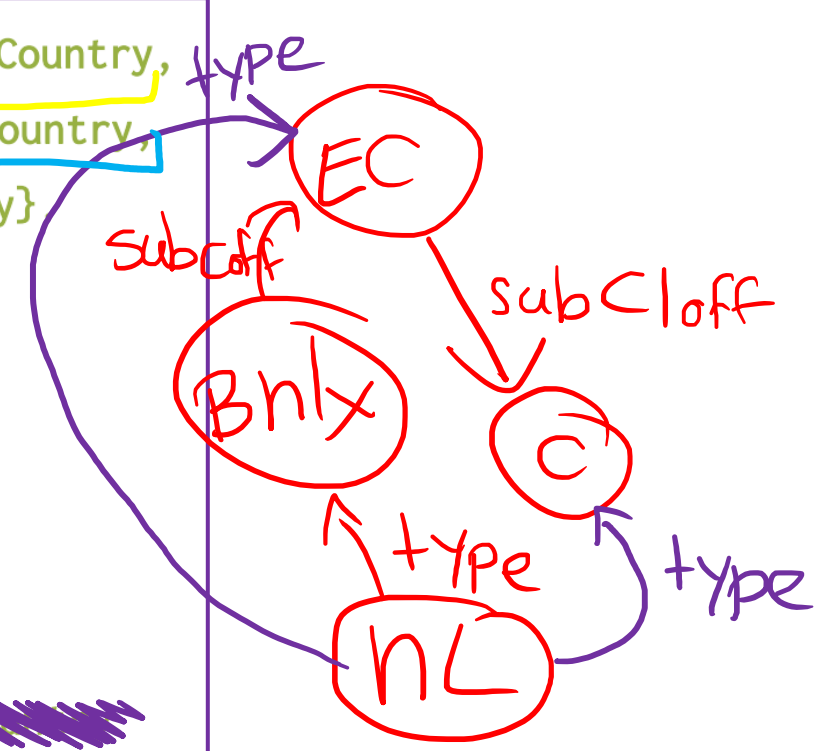
KG = {ex:EuropeanCountry rdfs:subClassOf ex:Country,
ex:BeneluxCountry rdfs:subClassOf ex:EuropeanCountry,
 ex:Netherlands rdf:type ex:BeneluxCountry}

5.	<u>s</u>	rdfs:subClassOf	<u>y</u>	$\Rightarrow$	s	rdfs:subClassOf	z
	y	rdfs:subClassOf	z				

10.	s	rdf:type	o				
	o	rdfs:subClassOf	t	$\Rightarrow$	s	rdf:type	t

This means that our little Knowledge Graph entails:

KG $\vdash_{\text{rdf}}$ ex:Netherlands rdf:type ex:Country



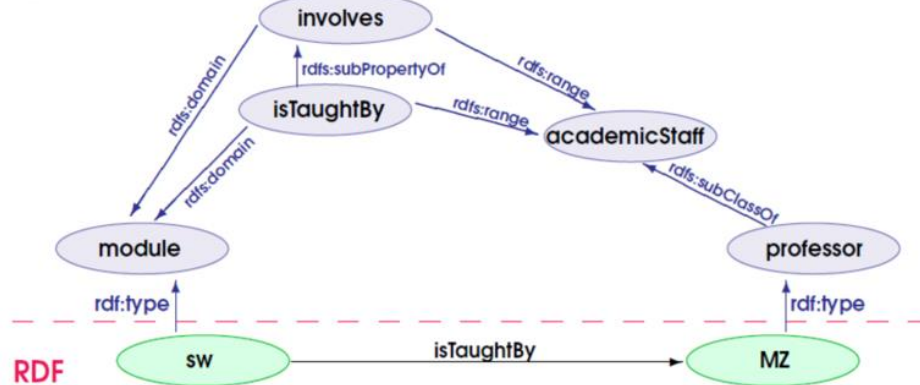
5: ex:BnlxC rdfs:subCl ex:country

10: ex:Netherlands rdf:type ex:country

Knowledge Graphs: Recap Question

Exercise 3: Knowledge Graphs: Write down the following Knowledge Graph as a set of triples.
Discuss whether there are any non-trivial triples that are entailed.

RDF Schema



RDF

Triple:
SW isTaughtBy MZ

MZ rdf:type professor
SW rdf:type module

prof rdfs:subClassOf academicStaff

rule 10: MZ rdf:type academicStaff

Rule 7:
TBY spo involves
SW TBY MZ
⇒ SW involves MZ

5. $s \text{ rdfs:subClassOf } y \implies s \text{ rdfs:subClassOf } z$
 $y \text{ rdfs:subClassOf } z$

6. $p \text{ rdfs:subPropertyOf } q \implies p \text{ rdfs:subPropertyOf } r$
 $q \text{ rdfs:subPropertyOf } r$

7. $s \text{ rdfs:subPropertyOf } o \implies s \text{ q } o$
 $p \text{ rdfs:subPropertyOf } q$

8. $s \text{ p } o \implies s \text{ rdf:type } x$
 $p \text{ rdfs:domain } x$

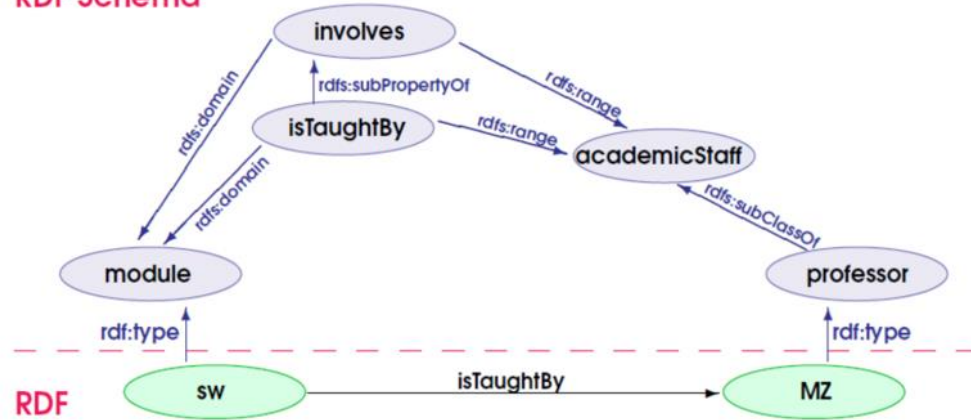
9. $s \text{ p } o \implies o \text{ rdf:type } x$
 $p \text{ rdfs:range } x$

10. $s \text{ rdf:type } o \implies s \text{ rdf:type } t$
 $o \text{ rdfs:subClassOf } t$

Recap Question: Triples Answer

Exercise 3: Knowledge Graphs: Write down the following Knowledge Graph as a set of triples. Discuss whether there are any non-trivial triples that are entailed.

RDF Schema



SUBJECT	PREDICATE	OBJECT
SW	rdfs:type	module
SW	isTaughtBy	MZ
MZ	rdfs:type	professor
isTaughtBy	rdfs:domain	module
isTaughtBy	rdfs:subPropertyOf	involves
isTaughtBy	rdfs:range	academicStaff
involves	rdfs:domain	module
involves	rdfs:range	academicStaff
professor	rdfs:subClassOf	academicStaff

Reasoning with RDFS rules: Recap Question

Exercise 3: Reasoning with RDFS rules: Suppose the following rule (only) is giving.

rdfs9: xxx rdfs:subClassOf yyy AND zzz rdf:type xxx $\rightarrow$ zzz rdf:type yyy .

and an RDFS graph:

{ (ex:A rdfs:subClassOf ex:D),
(ex:D rdfs:subClassOf ex:C),
(ex:B rdfs:subClassOf ex:D),
(ex:i1 rdf:type ex:A),
(ex:i2 rdf:type ex:B)}

Which triples can be derived?

$\Rightarrow$ ex:i1 rdf:type ex:D

$\Rightarrow$ ex:i2 rdf:type ex:D

Reasoning with RDFS rules: Recap Question

Consider the following knowledge graph, written in triple format.

```
ex:Charlie rdf:type ex:Person
ex:Charlie ex:paints ex:Wildflowers
ex:Adia ex:worksWith ex:Charlie
ex:Adia rdf:type ex:Sculptor
ex:paints rdfs:domain ex:Artist
ex:paints rdfs:range ex:Painting
ex:Sculptor rdfs:subClassOf ex:Artist
ex:worksWith rdfs:subPropertyOf ex:speaksTo
```

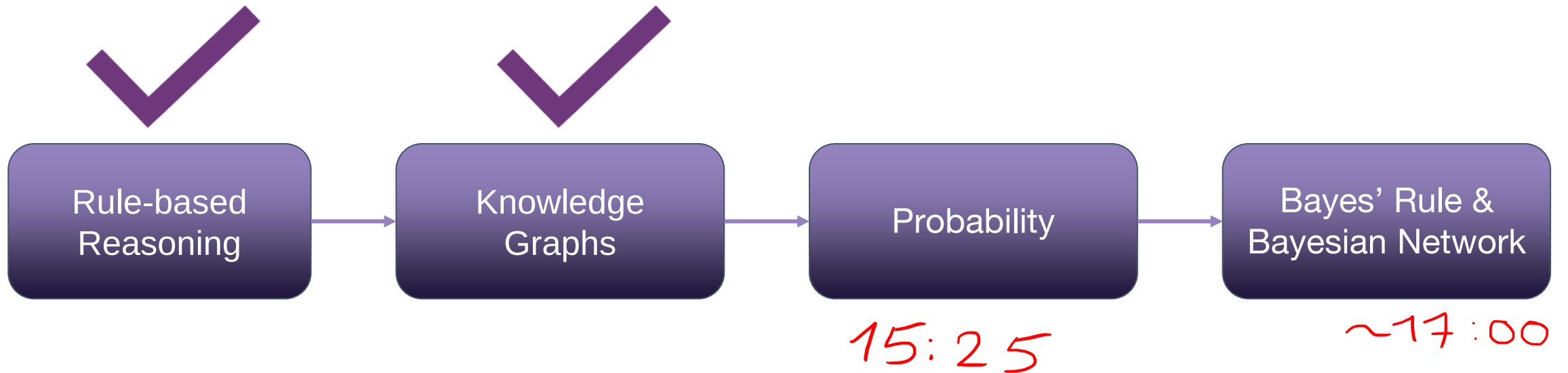
only
infer
from
knowledge
graph

Write down all triples that can be inferred using the rdfs inferencing rules.

5. $\begin{matrix} s & \text{rdfs:subClassOf} & y \\ y & \text{rdfs:subClassOf} & z \end{matrix} \Rightarrow s \text{ rdfs:subClassOf } z$
6. $\begin{matrix} p & \text{rdfs:subPropertyOf} & q \\ q & \text{rdfs:subPropertyOf} & r \end{matrix} \Rightarrow p \text{ rdfs:subPropertyOf } r$
7. $\begin{matrix} s & & o \\ p & \text{rdfs:subPropertyOf} & q \end{matrix} \Rightarrow s \quad q \quad o$
8. $\begin{matrix} s & & o \\ p & \text{rdfs:domain} & x \end{matrix} \Rightarrow s \quad \text{rdf:type} \quad x$
9. $\begin{matrix} s & & o \\ p & \text{rdfs:range} & x \end{matrix} \Rightarrow o \quad \text{rdf:type} \quad x$
10. $\begin{matrix} s & \text{rdf:type} & o \\ o & \text{rdfs:subClassOf} & t \end{matrix} \Rightarrow s \quad \text{rdf:type} \quad t$

- 7: ex:Adia ex:SpeaksTo ex:Charlie
- 9: ex:wildflowers rdf:type ex:painting
- 8: ex:Charlie rdf:type ex:Artist
- 10: ex:Adia rdf:type ex:Artist

Planning Day 1



Probability

Sample Space

Sample space: $\Omega = \{...\}$

It is the set of single outcomes of our experiment

die roll $\Omega = \{1, 2, 3, 4, 5, 6\}$

Can it also include pairs?

$\Rightarrow$ yes! 2 dice $\Omega = \{(1; 4), (2; 5), \dots (6; 6)\}$

Sample space examples



$$\Omega = \{\text{heads, tails}\}$$



$$\Omega = \{1, 2, 3, 4, 5, 6\}$$



$$\Omega = \{(1;1), (1;2), (2;1), (2;2), \dots, (6;6)\}$$

Subsets of the sample space, all possible events

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

What are all possible events of a die roll?

$$E = \{...\}$$

Roll an even number $E = \{2, 4, 6\}$

Roll a 3 $E = \{3\}$

Roll 1 or 5 $E = \{1, 5\}$

Roll 2 and 6 $E = \{ \} \leftrightarrow$ with 1 die

Events and Probability



$$E = \{\{\}, \{1\}, \{2\}, \{3\}, \dots, \{1, 2\}, \dots, \{1, 2, 3, 4, 5, 6\}\}$$

Using events to calculate probability: *denoted with capital P* 

$P(\{1, 2, 3\})$

probability that outcome < 4

$P(\{2, 4, 6\})$

probability that outcome is an even number

$P(\{6\})$

probability that outcome = 6

Probability Distribution

The probability distribution gives a value for all possible assignments, denoted with **P**

bold

When asked for $P(\dots)$, give your answer in a vector of probability values:

$\langle \dots = \text{True}, \dots = \text{False} \rangle$

sum to 1

The value of the probability vector should always lead to a sum of 1

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

sum to 1

Conditional Probability

$P(A|B)$ ^{“given”}

“The probability of A given B”

Conditional Probability Formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

Conditional Probability

$$\rightarrow \Omega = \{1, 2, 3, 4, 5, 6\}$$

What is $P(\{3\} | \text{uneven})$?

$$P(\{3\} \& \text{uneven}) = \frac{1}{6}$$

$$P(\text{uneven}) = \frac{3}{6} = \frac{1}{2}$$

$$P(\{3\} | \text{uneven}) = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

Inference by enumeration

$P(\text{Toothache}, \text{Catch}, \text{Cavity})$

	<i>toothache</i>		$\neg \text{toothache}$	
	<i>catch</i>	$\neg \text{catch}$	<i>catch</i>	$\neg \text{catch}$
<i>cavity</i>	.108	.012	.072	.008
$\neg \text{cavity}$	.016	.064	.144	.576

For any proposition ϕ , sum the atomic events where it is true: $P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$

What does this mean for $P(\text{cavity})$?

Sum where
cavity = true

$$P(\text{cavity}) = .108 + .012 + .072 + .008 = 0.2$$

$$P(\neg \text{cavity}) = 1 - 0.2 = 0.8$$

Recap Question: Probability Distribution

What is $P(B = \text{true})$?

$$0.03 + 0.25 + 0.07 + 0.33 = 0.68$$

What is $P(A \ \& \ B = \text{true})$?

$$0.07 + 0.33 = 0.4$$

What is $P(B \vee C = \text{true})$?

$$0.8 = 0.10 + 0.03 + 0.25 + 0.02 + 0.07 + 0.33$$

$$1 - 0.2 = 0.8 \text{ (same answer)} \quad \nabla \text{ Sum} = 1 \quad \nabla$$

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

Recap Question: Probability

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576



$$P(A|B) = \frac{P(A,B)}{P(B)}$$

Given this full joint probability distribution, calculate the following:

1. $P(\neg \text{toothache} \mid \text{cavity})$

$$P(\neg \text{tooth} \& \text{cavity}) = 0.072 + 0.008 = 0.08$$

$$P(\text{cavity}) = 0.2 \Rightarrow \frac{0.08}{0.2} = 0.4$$

2. $P(\text{Cavity})$

$P(\text{cavity}) = 0.2$
 $P(\neg \text{cavity}) = 0.8$
 $\Rightarrow \langle 0.2, 0.8 \rangle \Rightarrow \text{sum} = 1$

Recap Question: Probability

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

- $P(\text{toothache})$.
- $P(\text{Cavity})$.
- $P(\text{Toothache}|\text{cavity})$.
- $P(\text{Cavity}|\text{toothache} \vee \text{catch})$.

Bold P / Capital letter

P = give the probability vector

$\langle \dots = \text{true}, \dots = \text{false} \rangle$

$\hookrightarrow \nabla$ sum to 1

Capital letter = give value and $\neg$ value

$$\mathbf{P}(\underline{\text{Toothache}} | \text{catch}) = \frac{P(\underline{\text{tooth}} | \text{catch})^a}{P(\underline{\neg \text{tooth}} | \text{catch})^b} \Rightarrow \langle a, b \rangle$$

(In)dependent events

Step Plan:

① calculate
 $P(A|B)$

② calculate
 $P(A)$

③ check

$P(A|B) = P(A) \Rightarrow \text{indp. else, dep.}$

$$P(A|B) = P(A)$$

B has no effect on A
 A and B are independent



$$P(A|B) \neq P(A)$$

B has an effect on A
 A and B are dependent

Recap Question: (in)dependence

You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pair of events are (in)dependent.

- A: the sum is 5
- B: the red dice shows a 2

$$P(A|B) = P(\text{sum} = 5 | r = 2) \\ = 1/6$$

$$P(A) = P(\text{sum} = 5) \\ = 4/36 = 1/9$$

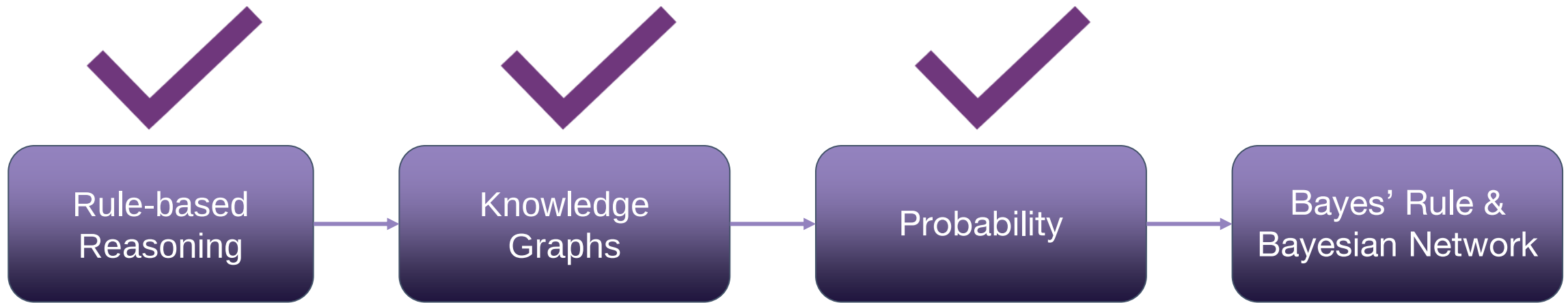
$$\Rightarrow 1+4, 4+1, 2+3, 3+2 \quad \text{sum} = 5$$

$$1/6 \neq 1/9 \\ \Rightarrow \text{dependent}$$

$$r = 2 \quad \text{---} \quad = 5 \\ g = 1, 2, 3, 4, 5, 6$$

$$6^2 = 36 \text{ events} \quad \left\{ \begin{array}{l} r = \textcircled{1}, \textcircled{2}, \textcircled{3}, \textcircled{4}, 5, 6 \\ g = \textcircled{1}, \textcircled{2}, \textcircled{3}, \textcircled{4}, 5, 6 \end{array} \right.$$

Planning Day 1



16:15
~ 17:00

Bayes' Rule & Bayesian Networks



$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Another way of computing conditional probability

By using Bayes' rule, we will get a more accurate probability value

Bayes Rule: Step Plan

1. Write down the probability that you want to find
2. Write down all the probability values that have been given in the question
3. Write down Bayes' Rule and discover how you can plug in the values to get to your final answer

Recap Question: Bayes' Rule

Given the following information:

- The probability that someone with a cold also has fever is 80%.
- The prior probability of having a cold is 0.0001.
- More or less 1 out of 1000 people in the population has fever.

I have fever. Calculate with Bayes' Rule the probability that I have cold.

$$P = (\text{cold} \mid \text{fever}) = ?$$

$$P(\text{fever} \mid \text{cold}) = 0.8$$

$$P(\text{cold}) = 0.0001$$

$$P(\text{fever}) = 0.001$$

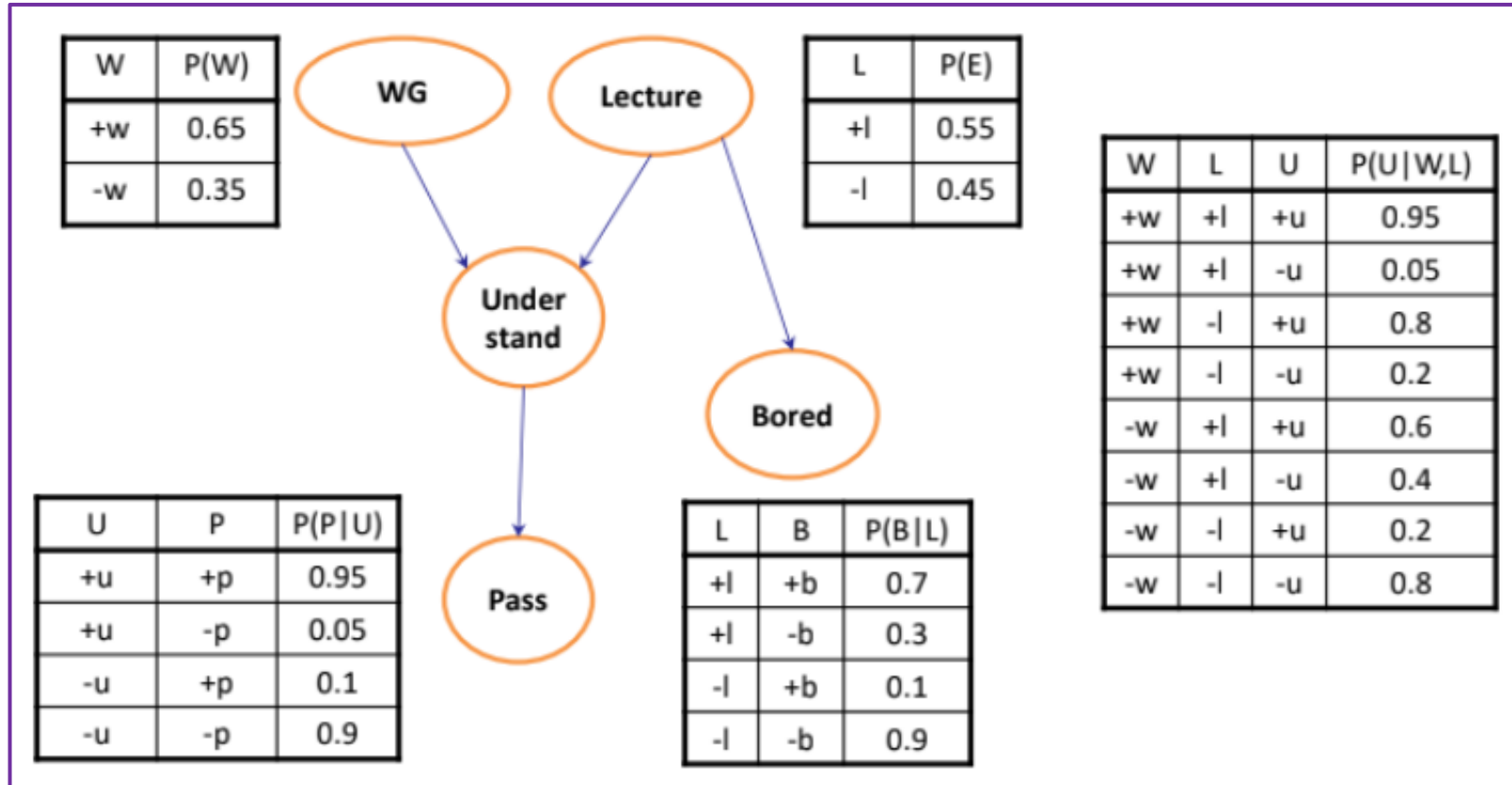
Prob I have cold given fever is 0.08

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

$$P = \frac{0.8 \cdot 0.0001}{0.001} =$$

$$0.08 \quad (8\%)$$

Bayesian Networks



Bayesian Networks: Step Plan

1. Write down the probability that we want to find
2. Is that probability dependent on some other variables?

No

→ Look up the value in the network

Yes

→ Write down all probabilities you need to calculate

3. Calculate probabilities with formula if needed:



$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(x_i))$$

4. Sum probabilities

$$P(+u, +wg) = P(+u | +wg) \cdot P(+wg)$$

Recap Question: Bayesian Networks

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

1
2
3
4

$$P(+a, +b, +e) = P(+a|+b, +e) \cdot P(+b) \cdot P(+e)$$

$$= 0.95 \cdot 0.001 \cdot 0.002$$

$$= 0.0000019 \text{ ①}$$

$$P(+a, +b, -e) = P(+a|+b, -e) \cdot P(+b) \cdot P(-e)$$

$$= 0.94 \cdot 0.001 \cdot 0.998$$

$$= 0.00093812 \text{ ②}$$

$$P(+a, -b, +e) = 0.999 \cdot 0.002 \cdot 0.29$$

$$= 0.00057942 \text{ ③}$$

$$P(+a, -b, -e) = 0.001 \cdot 0.999 \cdot 0.998$$

Given the Bayesian network, what is the probability that the alarm rings?

$$P(+a) = ? , \text{ dependent on } b, e$$

$$P(+a, -b, -e) = 0.000997002 \textcircled{4}$$

$P(+a)$ sum of all $\textcircled{4}$ we just calculated

$$\Rightarrow 0.002576442 = P(+a)$$

$$P(+a, -b, -e) = P(+a|-b, -e) \cdot P(-b) \cdot P(-e)$$

Recap Question: Bayesian Networks

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+a) \Rightarrow b, e$$

$$\textcircled{1} P(+j, +a) = P(+j|+a) \cdot \underbrace{P(+a)}_{\text{previously calculated}}$$

$$= 0.9 \cdot P(+a) = 0.0022648$$

$$\textcircled{2} P(+j, -a) = P(+j|-a) \cdot P(-a)$$

$$1 - P(+a) = P(-a)$$

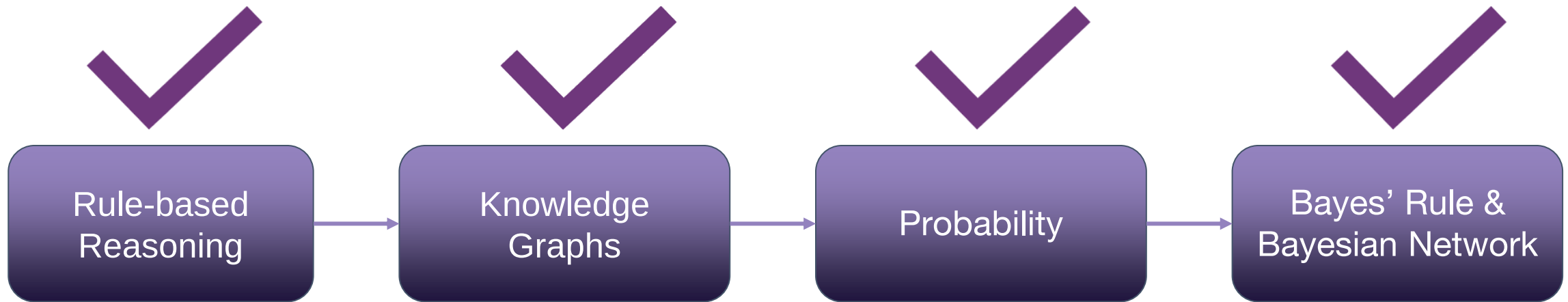
$$= 0.05 \cdot P(-a)$$

$$= 0.049874$$

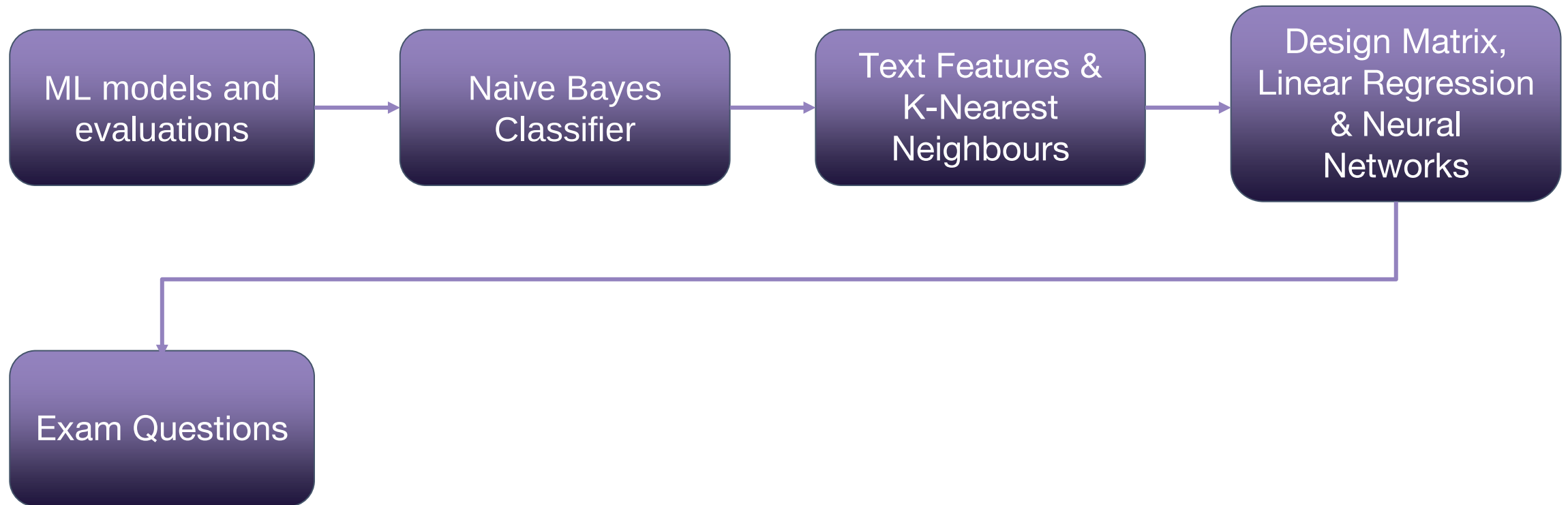
Given the Bayesian network, what is the probability that John calls?

$$P(+j) \Rightarrow \text{dep on } +a, -a \quad \Bigg| \quad \text{Sum } \textcircled{1} + \textcircled{2} = 0.05213898$$

Planning Day 1



Planning Day 2



Machine Learning: ML models and evaluations



Classification

From data to discrete classes.

Example:
Classify if this is considered as spam or not

Regression

Predicting a numeric value.

Example:
Stock market predictions.

Ranking

Comparing items.

Example:
Find similar images on the internet.

Collaborative Filtering

Recommending something.

Example:
Recommendation system for new movies.

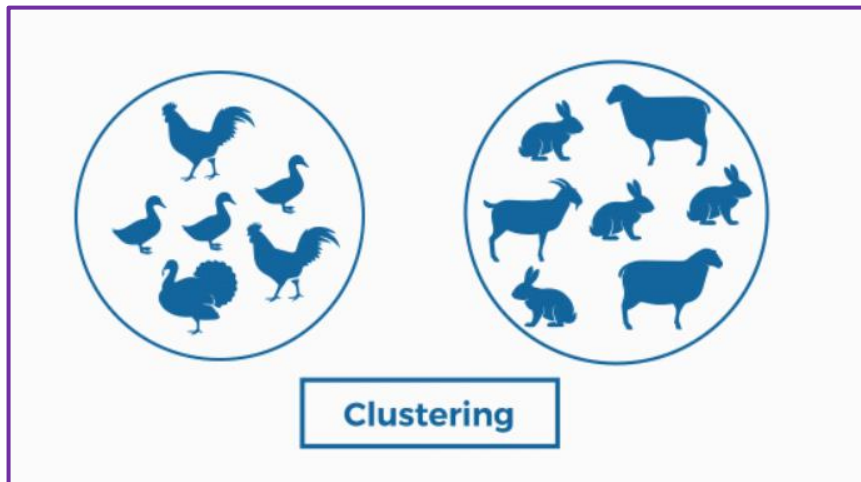
Clustering

Discovering structure in data and grouping similar items together.

Example:
Clustering web search results.

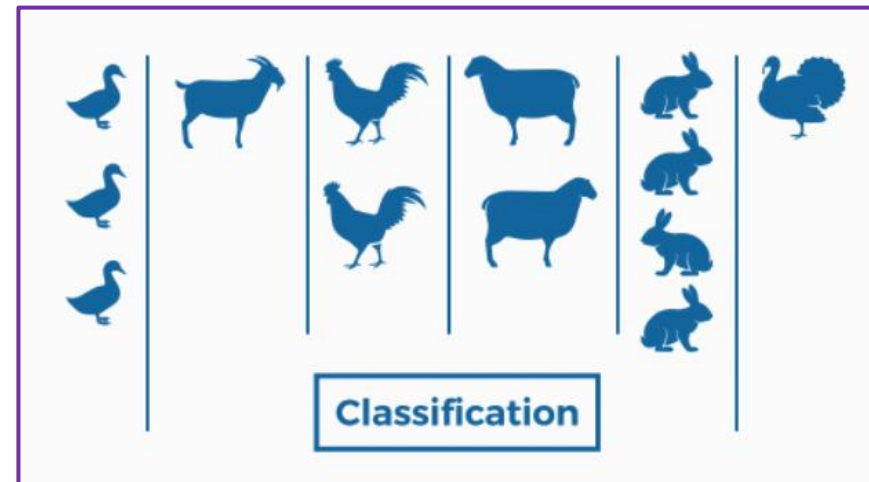
Classification vs Clustering

They both group data together, so what is the difference?



Clustering identifies **similarities**, which it groups together according to those characteristics. These groups are known as "**clusters**".

unsupervised



Classification uses **predefined classes** in which objects are assigned

supervised

Supervised vs. Unsupervised



Supervised

The program is trained and will be given the correct answer for each example.

Previous information is needed in order to function properly.

Unsupervised

The program is **not** trained and no correct answers will be given.

No previous information is needed in order to function properly.

Recap question

For the following questions, define which ML model needs to be used and if it is supervised or unsupervised:

1) A model that predicts the price of a house given certain features like size, how old the house is etc.

→ SUPERVISED

SUPERVISED, REGRESSION

2) We want to identify which cells are cancerous in the body.

CLUSTERING, UNSUPERVISED

3) We want to know whether a food contains certain allergens.

CLASSIFICATION, SUPERVISED

→ classes:
nuts
dairy
gluten

Precision vs Recall

Precision

$$TP / (TP + \text{FP})$$

Precision describes how *precise* the results were.

P ↑
R ↓

Recall

$$TP / (TP + \text{FN})$$

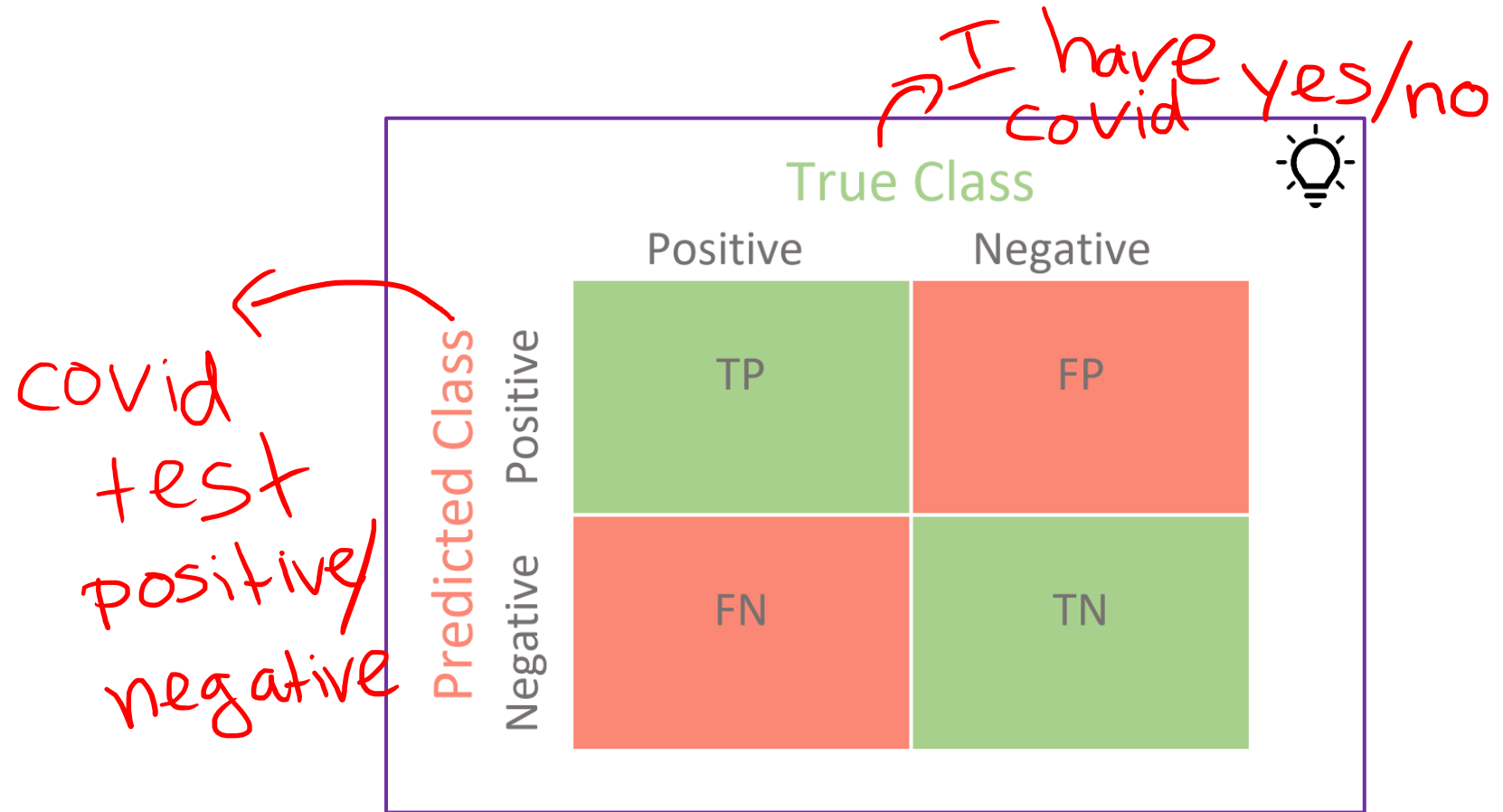
Recall describes the amount of actual *relevant* results.

R ↑
P ↓

Precision vs Recall

Used to evaluate
classification

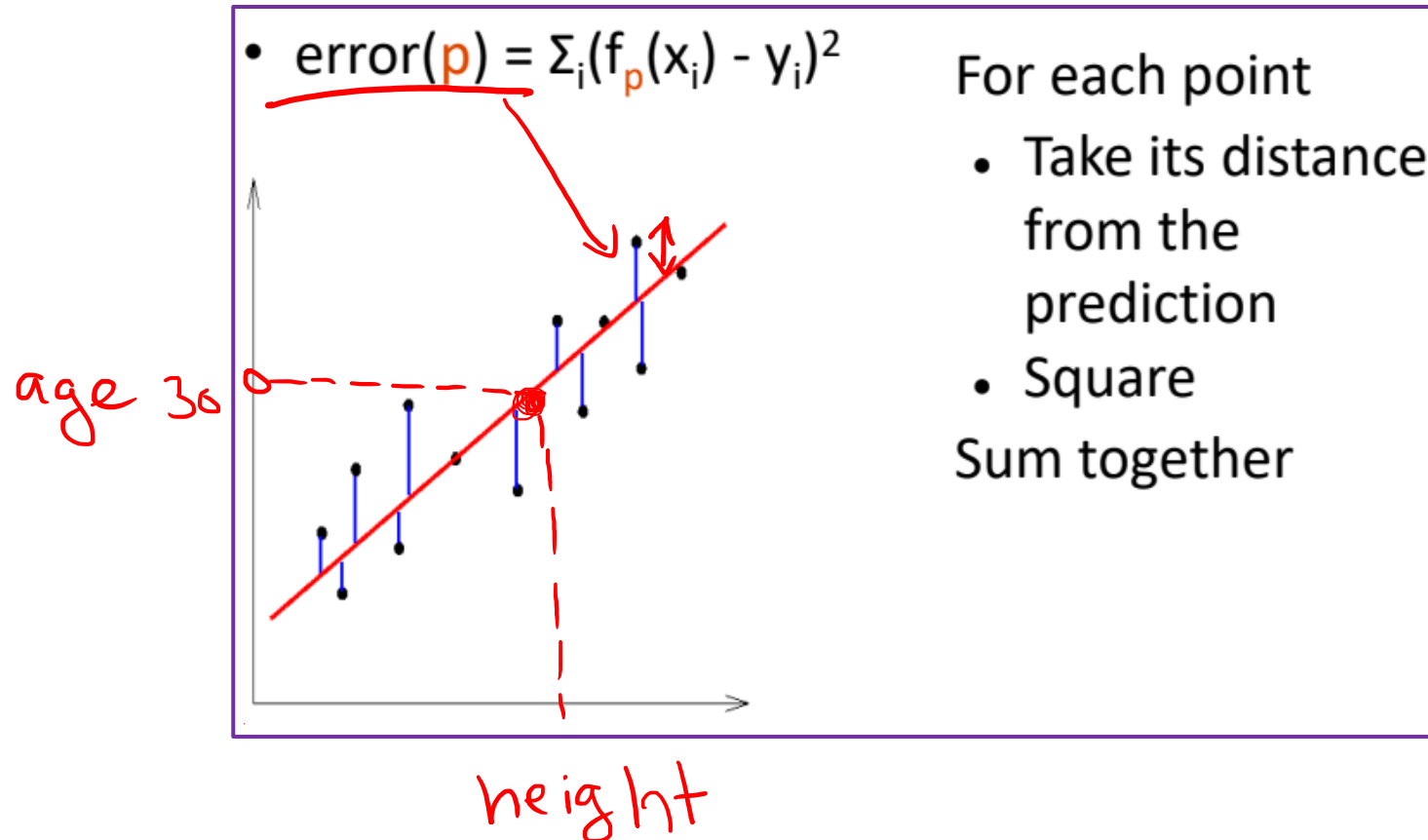
Data is depicted in a
confusion matrix:



Error value

Used to evaluate regression

↳ as low as possible
for better performance





Used to evaluate **recommendations** (collaborative filtering) and **ranking**

$$\text{Precision@}k = \frac{\text{true positives @}k}{(\text{true positives @}k) + (\text{false positives @}k)}$$

→ all options in total

$P@3$:

1.	✓
2.	✗
3.	✓
4.	
5.	

$$P@3 = \frac{2}{3}$$

Recap question: Precision vs Recall

System B got the following scores:

	actual +	actual -
predicted +	207 TP	50 FP
predicted -	3 FN	125 TN

Now, imagine the task was to predict whether a patient might possibly have cancer and after a positive test (it looks like the person has cancer), the person would be further investigated.

Calculate the precision and recall for this confusion matrix and conclude whether precision or recall is more important here.

Formulas:

$$P = TP / (TP + FP)$$

$$R = TP / (TP + FN)$$

$$P = 207 / (207 + 50) = 0.805 / 0.81$$

$$R = 207 / (207 + 3) = 0.99$$

Recall more important in order to min. FN?

Recap question: Error value

Exercise 3: Regression Errors

Consider a regression function $f(x) = x$ that was trained on some training set, and that is now evaluated with the following test-set of datapoints $\{(0,1), (1,2), (2,2.5), (3,2), (0,0), (1,1), (2,1), (3,3)\}$.

3a) What is the error value for this function?

$$f(x) = 2x, f(x) = -x$$

$$\text{error}(p) = \sum_i (f_p(x_i) - y_i)^2$$

$$f(x) = x$$

$$\Rightarrow \sum_i (x_i - y_i)^2$$

$$\text{ERROR}(p) = \sum_i (f_p(x_i) - y_i)^2$$

$$(0,1) \Rightarrow (0-1)^2 = 1 \quad (2-2.5)^2 = 0.25$$

$$(1-2) \Rightarrow (1-2)^2 = 1 \quad (3-2)^2 = 1$$

$$(0-0)^2 = 0 \quad (1-1)^2 = 0 \quad (2-1)^2 = 1 \quad (3-3)^2 = 0$$

$$\text{sum} = 1 + 1 + 1 + 1 + 0.25 = 4.25$$

Recap question: Precision at k

Exercise 4: Ranking Errors

Consider the following output of an algorithm written to retrieve Plays by Shakespeare:

- FP 1. Harry Potter and the Prisoner of Azkaban
- TP { 2. A Midsummer Night's Dream
- TP { 3. King Lear
- FP 4. Umbrella Academy
- TP { 5. Macbeth
- TP { 6. Much Ado About Nothing
- TP { 7. Romeo and Julia
- TP { 8. The Tempest
- FP 9. The Spring
- FP 10. Artificial Intelligence, a Modern Approach

What is the precision at 5, 7 and 10 of this ranking?

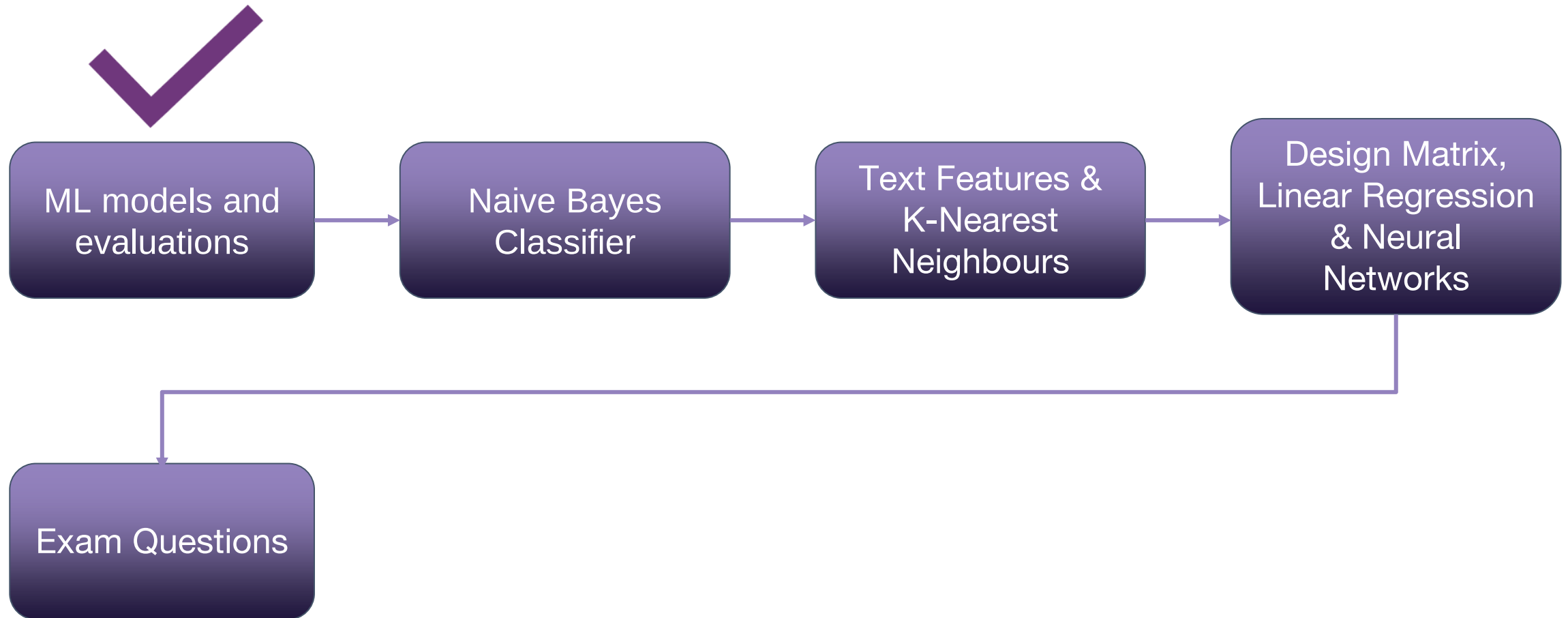
$$\frac{TP@k}{TP@k + FP@k} = P@k$$

$$P@5 = 0.6 \Rightarrow \frac{3}{5}$$

$$P@7 = \frac{5}{7}$$

$$P@10 = \frac{6}{10} = \frac{3}{5}$$

Planning Day 2



Machine Learning: Naive Bayes Classifier

Naive Bayes

Input

Data which belong to certain classes

One set of data an unknown class

Output

We want to calculate the probability that the data belongs to a certain class using this formula:

$$P(c | X) = P(x_1 | c) \times P(x_2 | c) \times \cdots \times P(x_n | c) \times P(c)$$

• why is it called naive?
=> assumes that all features are conditionally independent

Naive Bayes: Step Plan

1. Decide what the **main features** of the data (usually provided in the question)
2. Count the **occurences** of the words in all of the data
 - You will find which words are in the unknown data
3. Define the **probabilities** of this data belonging to one class or the other and calculate those probabilities
 - Use Laplace smoothing if needed $\rightarrow 0$ becomes 0.01 (small prob.)

Prevents that outcome = 0
4. Apply **naive bayes** to find the class with maximal probability

Recap question: Naive Bayes

Exercise 5: Bayesian Learning for text classification.

Suppose there are 5 documents about two courses “Intelligent Systems (IS)” and “Distributed Systems (DS)”:

IS D1: Logics and intelligent systems are very important in the course

DS D2: Intelligent algorithms are critical to solve problems in distributed environments

IS D3: Intelligent algorithms and logic are crucial

IS D4: Distributed systems are Intelligent

DS D5: Distributed systems need good algorithms

D1, D3 and D4 are about the Intelligent Systems class, D2 and D5 about Distributed Systems.

Use the (Boolean) features **Logic, Intelligent, Systems, Distributed and Algorithms** to predict whether the document D6 “Distributed algorithms can help reasoning in logic” is about the IS class, or DS.

Do this by making a naive Bayes classifier.¹

T, F ←

Recap question: Naive Bayes

C ; IS or DS

	Logic	Intelligent	Systems	Distributed	Algorithms	Course
D1	T	T	T	F	F	IS
D2	F	T	F	T	T	DS
D3	T	T	F	F	T	IS
D4	F	T	T	T	F	IS
D5	F	F	T	T	T	DS
D6	T	F	F	T	T	?

x_1 x_2 x_3 x_4 x_5

$$P(IS|D6) = ? , P(DS|D6) = ?$$

$$P(c|X) = P(x_1|c) \times P(x_2|c) \times \dots \times P(x_n|c) \times P(c)$$

$$P(DS|D6) = P(L|DS) \cdot P(\neg I|DS) \cdot \dots \cdot P(DS)$$

$$P(\underline{IS}|D6) = P(L|IS) \cdot P(\neg I|IS) \cdot (\neg S|IS) \cdot P(D|IS) \cdot P(A|IS) \cdot P(IS)$$

Recap question: Naive Bayes

	X = IS	X = DS
$P(L X)$	$2/3$	0.01
$P(\neg L X)$	0.01	$1/2$
$P(S X)$	$1/3$	$1/2$
$P(D X)$	$1/3$	1
$P(A X)$	$1/3$	1

↳ belong to D6

$$P(IS) = 3/5$$

$$P(DS) = 2/5$$

conclusion:

$$P(IS | D6) < P(DS | D6)$$

so D6 belongs to DS

$$\begin{aligned} P(IS | D6) &= \frac{2}{3} \cdot 0.01 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{3}{5} \\ &= 0.00015 \end{aligned}$$

$$\begin{aligned} P(DS | D6) &= 0.01 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 1 \cdot 1 \cdot \frac{2}{5} \\ &= 0.01 \end{aligned}$$

Recap question 2: Naive Bayes

There is a clear correlation between the weather, whether the NS operates smoothly and my level of frustration. The following table summarises some observations on five different days:

	Cold	Trains <i>Cancelled</i>	Snow	Wind	Classification: Frustrated
D1	Yes	No	No	Yes	No
D2	Yes	Yes	Yes	No	Yes
D3	No	Yes	No	Yes	Yes
D4	No	No	No	No	No
D5	Yes	Yes	No	Yes	No

D6 no no yes yes ?

Apply Naive Bayesian learning with a naïve smoothing of 0.01 on this data to predict whether I will be frustrated on a day where: it is not cold and no train is cancelled although it is windy and it snows.

conclusion:

$$P(\neg F | D6) > P(F | D6)$$

so I will likely not be frustrated on day 6

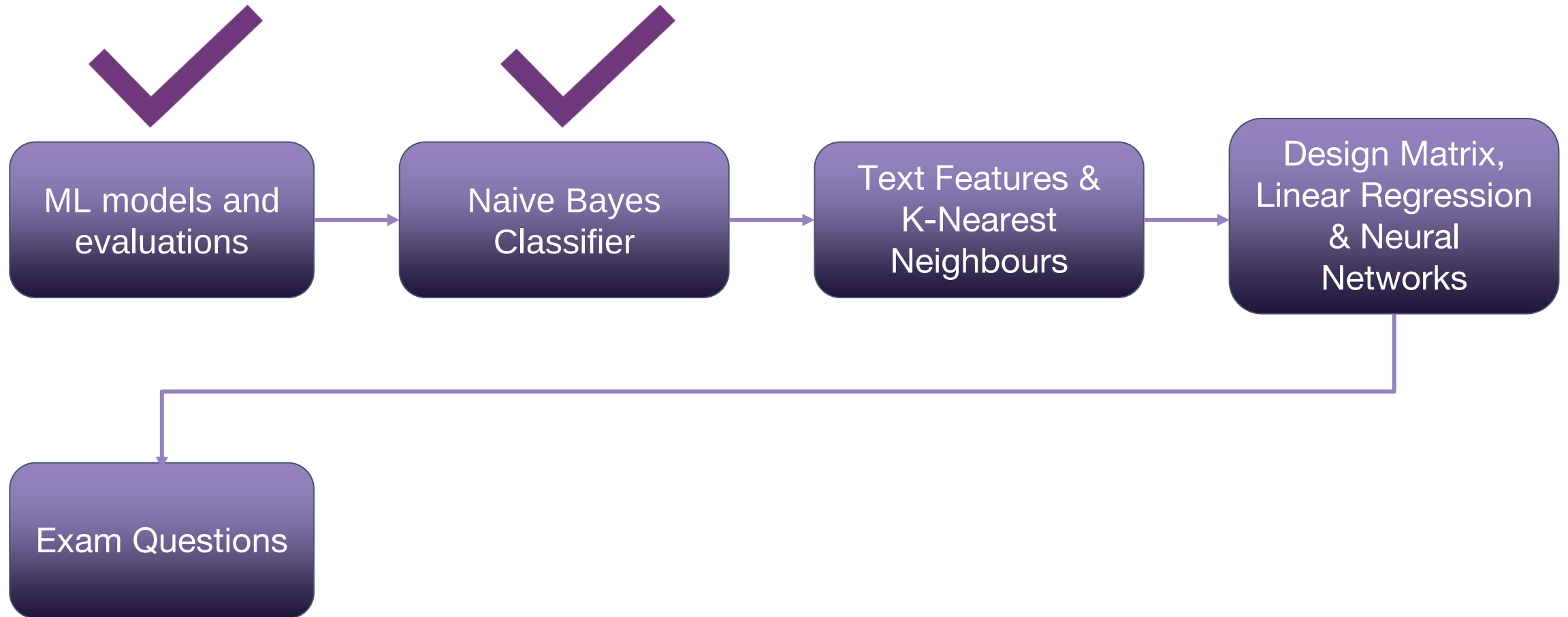
$$P(F | D6) = P(\neg C | F) \cdot P(\neg T | F) \cdot P(S | F) \cdot P(W | F) \cdot P(F)$$

$$P(\neg F | D6) = P(\neg C | \neg F) \cdot P(\neg T | \neg F) \cdot P(S | \neg F) \cdot P(W | \neg F) \cdot P(\neg F)$$

$$P(F | D6) = \frac{1}{2} \cdot 0.01 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{5} = 0.0005$$

$$P(\neg F | D6) = \frac{1}{3} \cdot \frac{2}{3} \cdot 0.01 \cdot \frac{2}{3} \cdot \frac{3}{5} = 0.00089$$

Planning Day 2



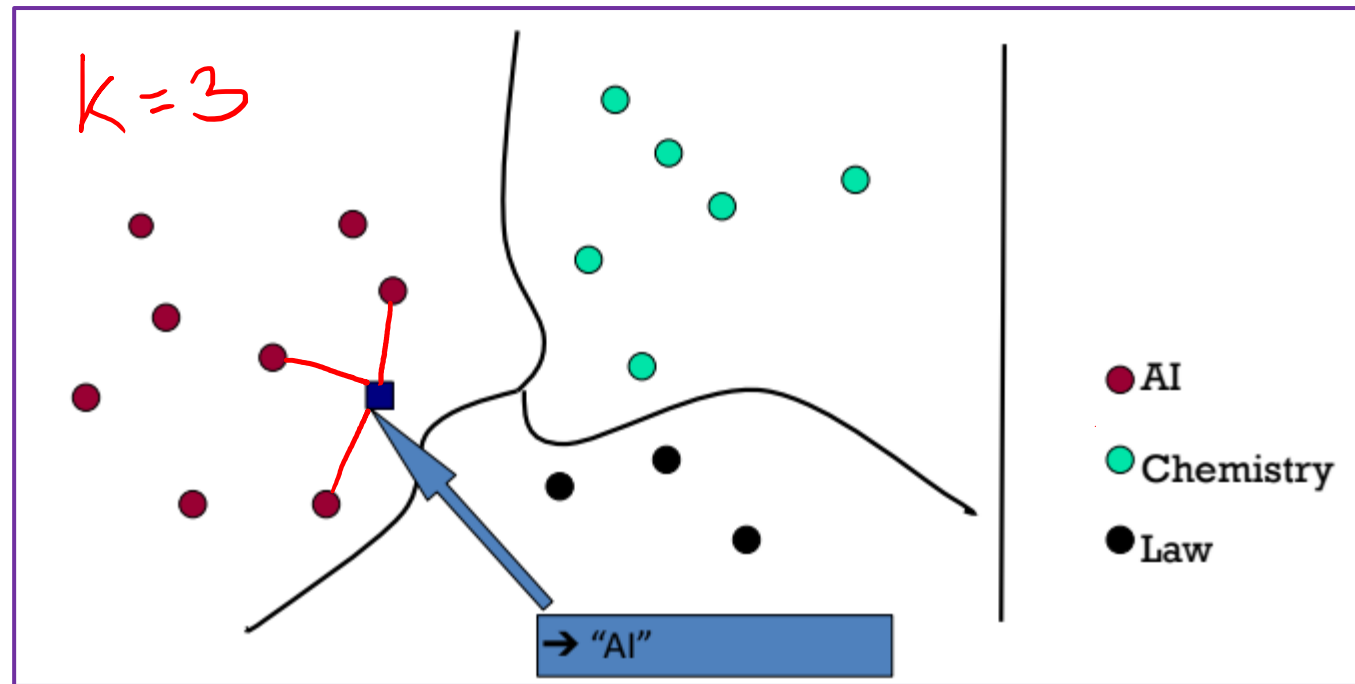
Machine Learning: Text Features & K-Nearest Neighbours

K-NN Classification

Used to classify unlabeled data

We want to find the most similar case and copy the label:

$k=2$
 $k=5$



Text Features

Two texts are similar if their constructed feature vectors are similar

$$\begin{matrix} & 1 & 2 & 3 & 4 & 5 \\ \langle & 1 & 1 & 0 & 0 & 1 \rangle \end{matrix}$$

How can we measure this?

$$\langle 0, 1, 0, 1, 1 \rangle$$

- Euclidean distance
- Hamming Distance
- Cosine Similarity

K-NN: Euclidean Distance

$$q_i \neq p_i$$

$$(1-0)^2 = 1$$

$$(0-1)^2 = 1$$

$$\sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

$$q_i = p_i$$

$$(0-0)^2 = 0$$

$$(1-1)^2 = 0$$

Smaller value $\rightarrow$ shorter 'distance' $\rightarrow$ closer neighbour

$\Rightarrow$ smallest values $\Rightarrow$ nearest neighbours

$\Rightarrow$ sum the amount of times where $q_i \neq p_i$, take $\sqrt{\quad}$

$$1 + 1 = 2 \Rightarrow \sqrt{2}$$

A: 0, 1, 0, 1, 1, 0
B: 1, 0, 0, 1, 1, 0

K-NN: Hamming Distance

$$\text{Hamming distance} = \frac{\text{\#bits different}}{\text{\#bits}^{\text{total}}}$$

Smaller value → shorter 'distance' → closer neighbour

0110010
0100110



*Differs in 2 positions so
Hamming distance is $\frac{2}{7}$*

K-NN: Cosine Similarity



$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}},$$

→ sum amount of times where A=1 and B=1

$$\begin{array}{l} 1 \cdot 1 = 1 \\ 0 \cdot 1 = 0 \\ 1 \cdot 0 = 0 \\ 0 \cdot 0 = 0 \end{array}$$

where A=1

where B=1

Calculates similarity, so bigger value → more similarity
→ closer neighbour

→ biggest values

Preprocessing Steps: Text Features

Before we can construct the feature vectors and use the formulas, we have to take some preprocessing steps:

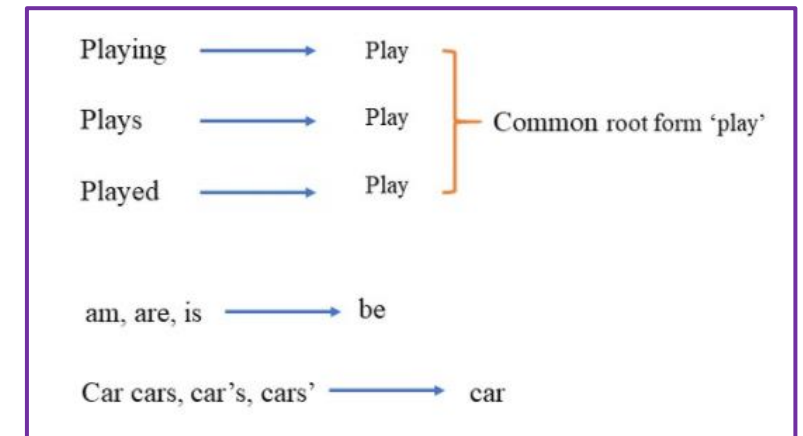
1. Lowercase text

2. Remove punctuation

3. Stemming

4. Remove stop words (usually given which ones)

Stemming:



Recap Question: Text Features

Exercise 6: Text Features

You are given the following text documents (one sentence is a document for this exercise).

1. A cat sleeps near the field. label: passive
2. The cow walks in the field. label: active
3. The field is full of poppies. label: passive
4. The cats and the dogs run in the field. label: active
5. The cow is mad because the cats and the dogs walked all over the poppies. label: active

Q1: convert each of these sentences into feature vectors. Use the following preprocessing steps:

- Lowercase text
- Remove punctuation
- Stemming (you can do this intuitively)
- Remove stop words (remove these words: a, the, of, in, all, because, over, on)
- you do not need to count occurrences as all of them should be one in this example (no need to do tf-idf, but you can if you want to, but only do that if you are solving this with a computer program and not on the paper)

Tip: create a table to write these vectors down, like this fictive example for the words {table, chair, desk, lamp}.

Table	Chair	Desk	lamp
1	1	0	0
0	1	0	1

Recap Question: Text Features

1: {cat, sleep, near, field}

2: {cow, walk, field}

3: {field, full, poppy, is}

4: {cat, and, dog, run, field}

5: {cow, is, mad, cat, and, dog, walk, poppy}

Recap Question: Text Features

Cat	Sleep	Near	Field	Cow	Walk	Dog	Is	Full	Poppy	And	Run	Mad
1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	1	1	0	0	0
1	0	0	1	0	0	1	0	0	0	1	1	0
1	0	0	0	1	1	1	1	0	1	1	0	1

Recap Question: K-NN for Text Classification

→ 2 nearest neighbours

You are creating a k-NN classifier with k=2 and using Euclidean distance as a metric.
To which class does the following sentence belong? Active or passive?

A = The dog walks in the field.

A: {dog, walk, field}

Cat	Sleep	Near	Field	Cow	Walk	Dog	Is	Full	Poppy	And	Run	Mad
0	0	0	1	0	1	1	0	0	0	0	0	0

Comparing Tables

	Cat	Sleep	Near	Field	Cow	Walk	Dog	Is	Full	Poppy	And	Run	Mad	label:
1	1	1	1	1	0	0	0	0	0	0	0	0	0	passive
2	0	0	0	1	1	1	0	0	0	0	0	0	0	<u>active</u>
3	0	0	0	1	0	0	0	1	1	1	0	0	0	passive
4	1	0	0	1	0	0	1	0	0	0	1	1	0	<u>active</u>
5	1	0	0	0	1	1	1	1	0	1	1	0	1	active

	Cat	Sleep	Near	Field	Cow	Walk	Dog	Is	Full	Poppy	And	Run	Mad	label:
A	0	0	0	1	0	1	1	0	0	0	0	0	0	?

Recap Question: K-NN for Text Classification

	1	2	3	4	5
A	5	2	5	4	7

	1	2	3	4	5
A	$\sqrt{5}$	$\sqrt{2}$	$\sqrt{5}$	$\sqrt{4}$	$\sqrt{7}$



$$\sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

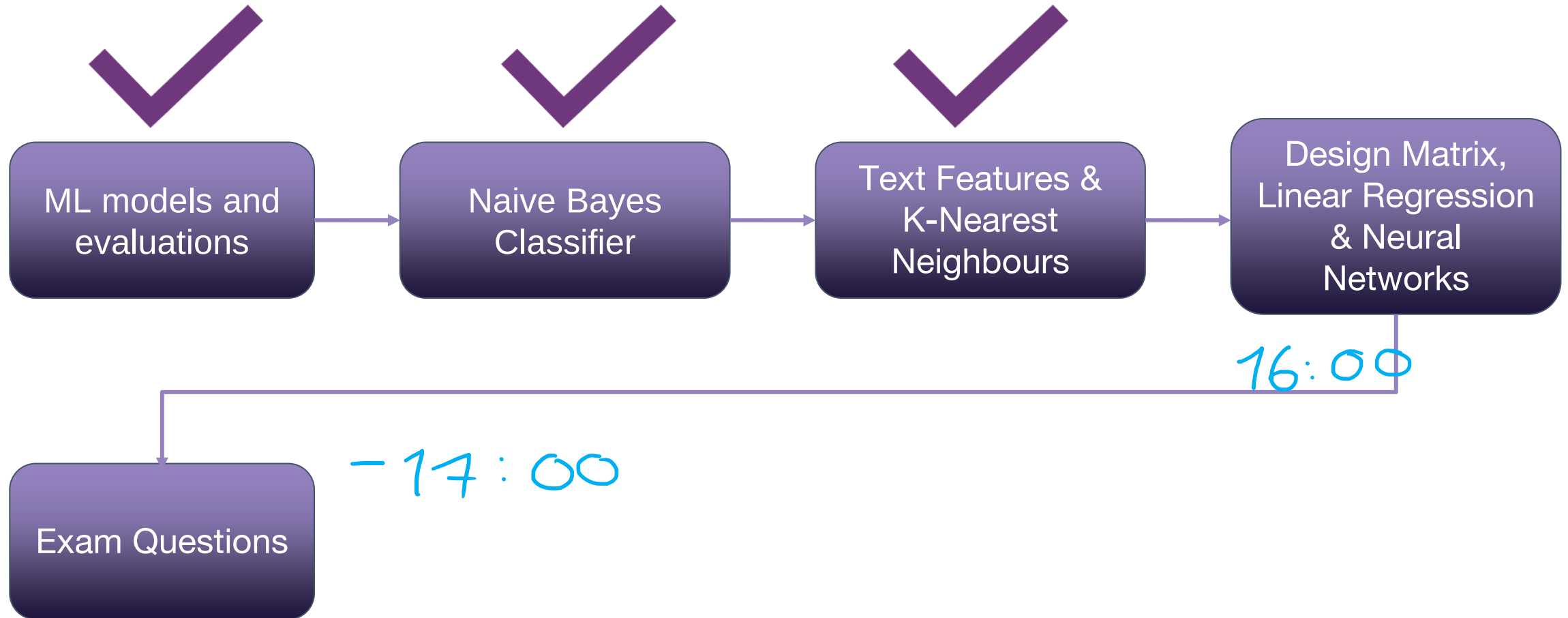
what if both
active, passive:
⑦ nearest neighbour

$k=2$

Smallest distance, so sentence
2 and 4 are the nn.
active active

$\Rightarrow$ sentence A belongs to active

Planning Day 2

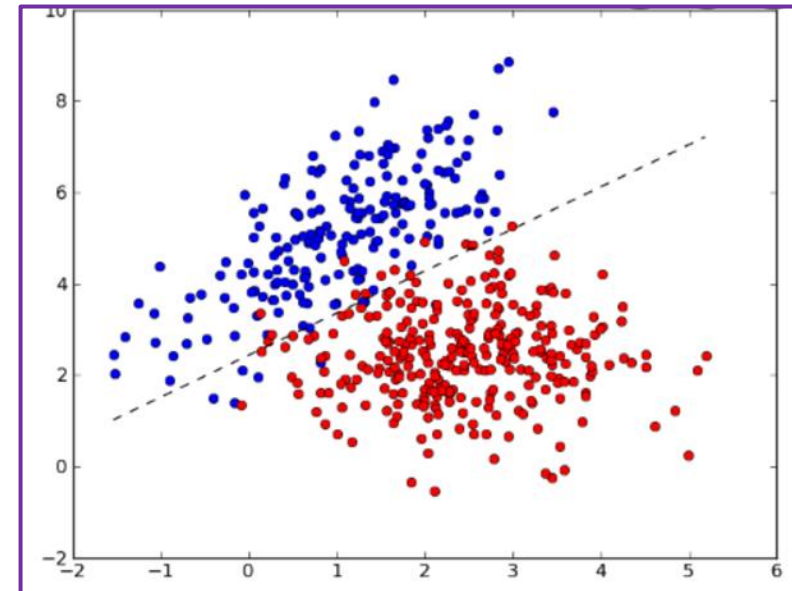
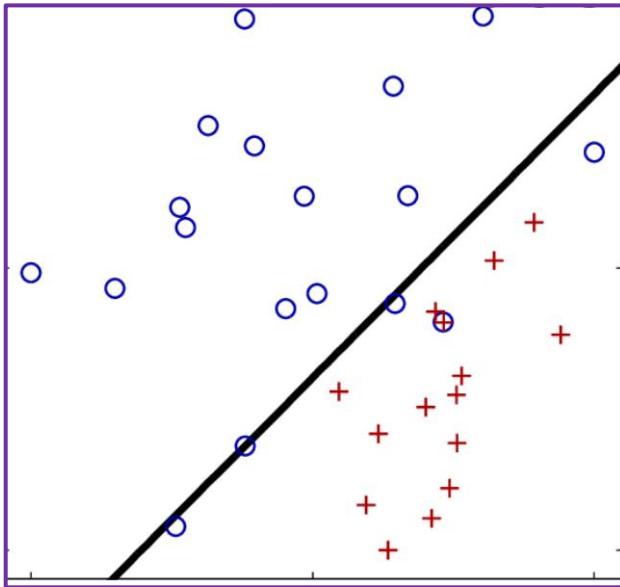


Machine Learning: Design Matrix, Linear Regression & Neural Networks

Design Matrix

With classification data, you always try to group the same kind of data together

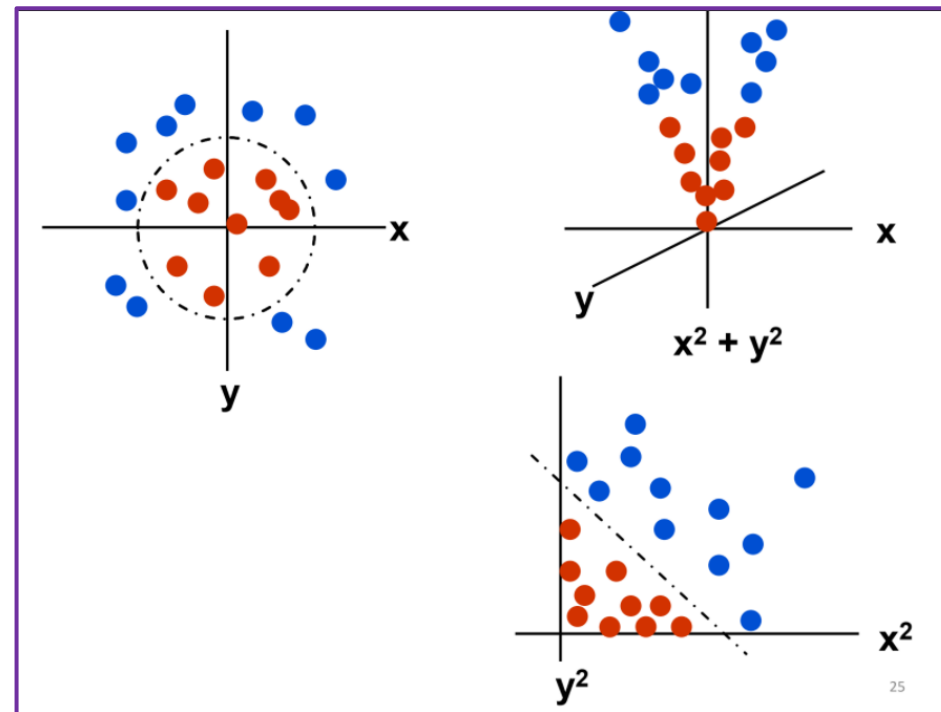
You can do this by using a linear model:



Design Matrix

What if you are not able to classify them with a linear model?

→ You will have to modify the features such that a linear model will fit



Design Matrix: Step Plan

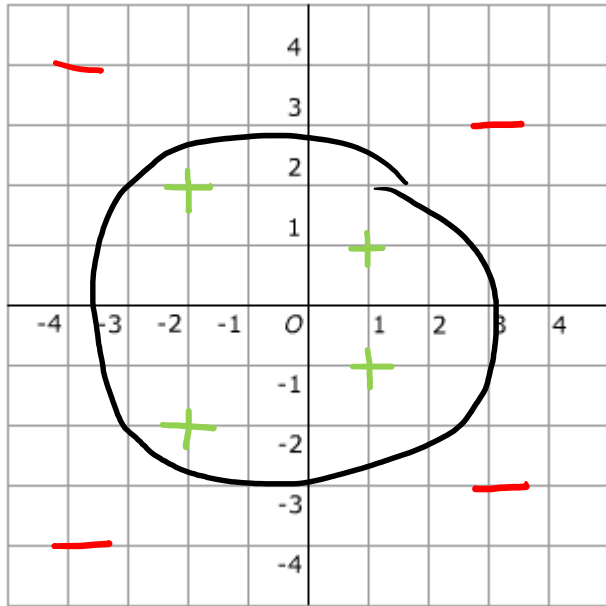
1. Draw the coordinate system of the original feature set
2. Provide a design matrix with new features and draw it
3. Define a linear separator that fits

$$y = ax + b$$

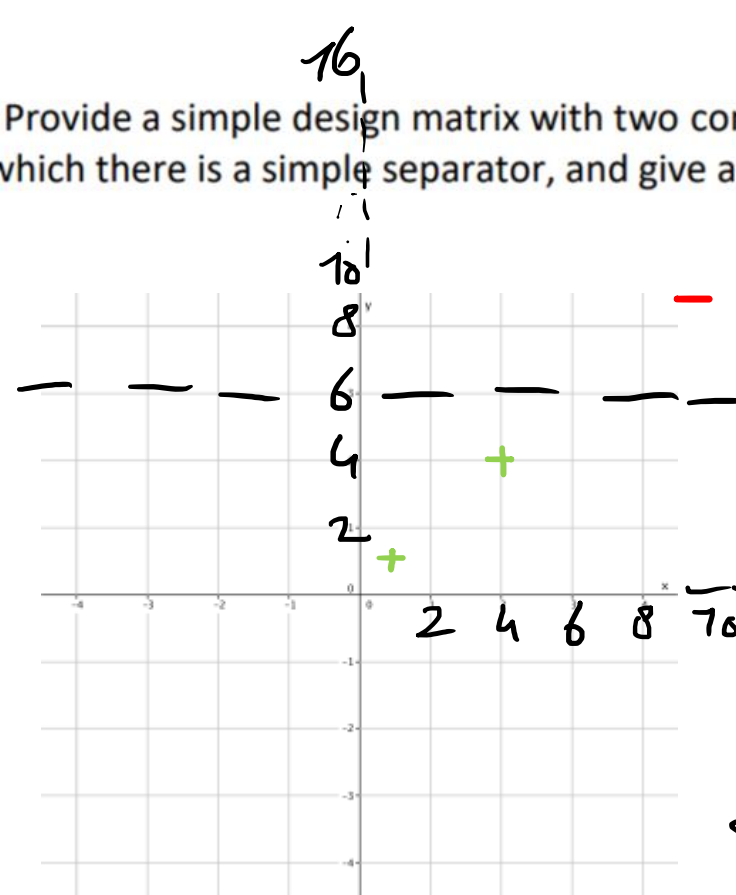
Recap question: Design Matrix

Consider the following training set of positive examples $\{(1,1), (1,-1), (-2,2), (-2,-2)\}$ and negative examples $\{(3,3), (3,-3), (-4,4), (-4,-4)\}$. Draw the feature space and try to fit a linear separator between the classes.

What can you do to rectify the problem? Provide a simple design matrix with two combined features instead of the original ones for which there is a simple separator, and give an example for such a linear separator.



$\Rightarrow$ no linear separator



$$x = x^2$$

$$y = y^2$$

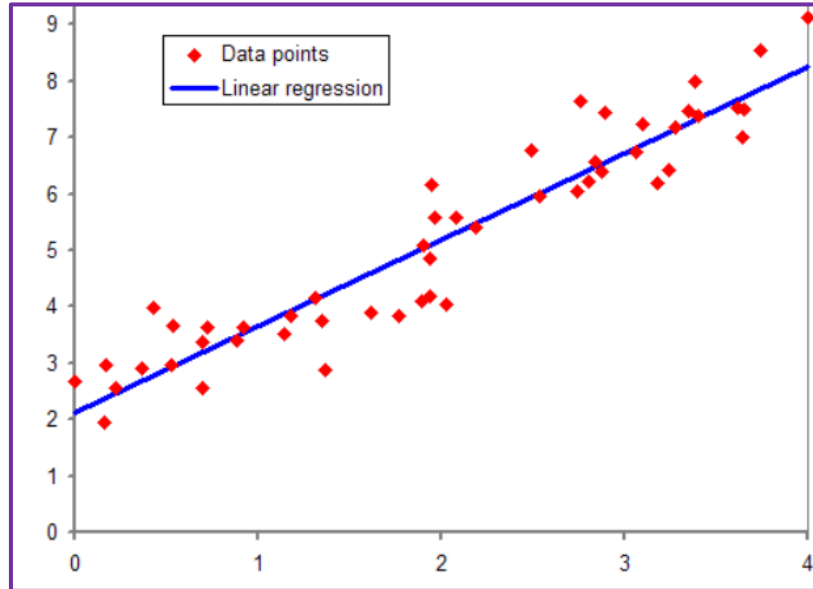
$$\text{neg} = \{(9,9), (16,16)\}$$

$$\text{pos} = \{(1,1), (4,4)\}$$

$$\text{separator: } f(x) = 6$$

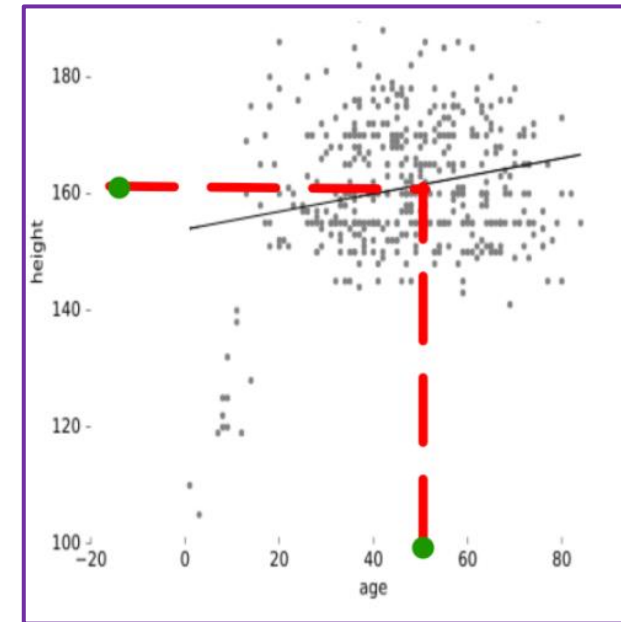
$$y = ax + b$$

Linear Regression



What was regression again?

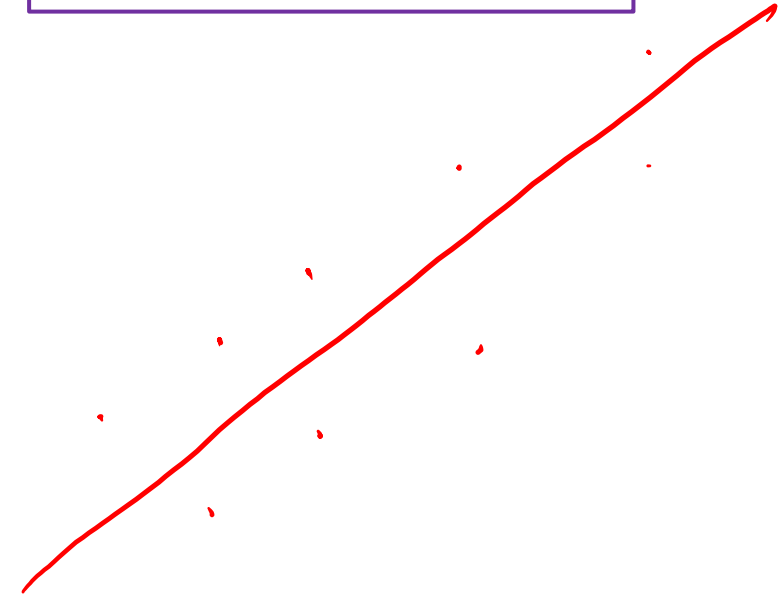
Linear regression: finding a linear function for a number of values so we can make more predictions



Linear Regression: Step Plan

1. Calculate the (squared) error function of each value
2. Calculate the total error function based on those values
3. Search for the minimal function of this value by:
 - a. Calculating the derivation of the function
 - b. Setting this derivation to 0 and solving the equation
4. The value you will find gives you the function that will be the optimal linear function that fits your data

$$\text{error}(\mathbf{p}) = \sum_i (f_{\mathbf{p}}(x_i) - y_i)^2$$



Recap Question: Linear Regression

Here we are going to practise fitting a linear function to a number of training sets. Suppose we have three pairs of values (1,1), (2,3) and (3,2) for two features. As discussed in the lecture, according to the ML recipe, the results depend on the chosen model class. Let us calculate the optimal function according to a number of different classes.

2a) Let us start with the most simple model class to fit our training examples, a constant line. This means that we are looking at a function $f(x) = b$, which is a horizontal line in the coordinate system. What is the result of learning the optimal function $f(x)$ from the given data? Hint: we are looking for a value for b .

$$\text{error}(p) = \sum_i (f_p(x_i) - y_i)^2$$

$$f(x) = b$$

$$f(x) = b$$

$$\sum_i (b - y_i)^2$$

$$\begin{aligned} \textcircled{1} \quad (1,1) &\rightarrow (b-1)^2 \xrightarrow{\textcircled{2}} (b-1)(b-1) = b^2 - 2b + 1 \\ (2,3) &\rightarrow (b-3)^2 \rightarrow (b-3)(b-3) = b^2 - 6b + 9 \\ (3,2) &\rightarrow (b-2)^2 \rightarrow (b-2)(b-2) = b^2 - 4b + 4 \end{aligned} \left. \vphantom{\begin{aligned} \textcircled{1} \quad (1,1) &\rightarrow (b-1)^2 \xrightarrow{\textcircled{2}} (b-1)(b-1) = b^2 - 2b + 1 \\ (2,3) &\rightarrow (b-3)^2 \rightarrow (b-3)(b-3) = b^2 - 6b + 9 \\ (3,2) &\rightarrow (b-2)^2 \rightarrow (b-2)(b-2) = b^2 - 4b + 4 \end{aligned}} \right\} \Rightarrow 3b^2 - 12b + 14$$

$$\textcircled{3} \quad 3b^2 - 12b + 14 \quad \begin{aligned} &\text{a: } \text{error}' = 6b - 12 \\ &\text{b: } 6b - 12 = 0 \quad \begin{aligned} &6b = 12 \quad \div 6 \quad b = 2 \end{aligned} \end{aligned}$$

$$\textcircled{4} \quad f(x) = b \Rightarrow f(x) = 2$$

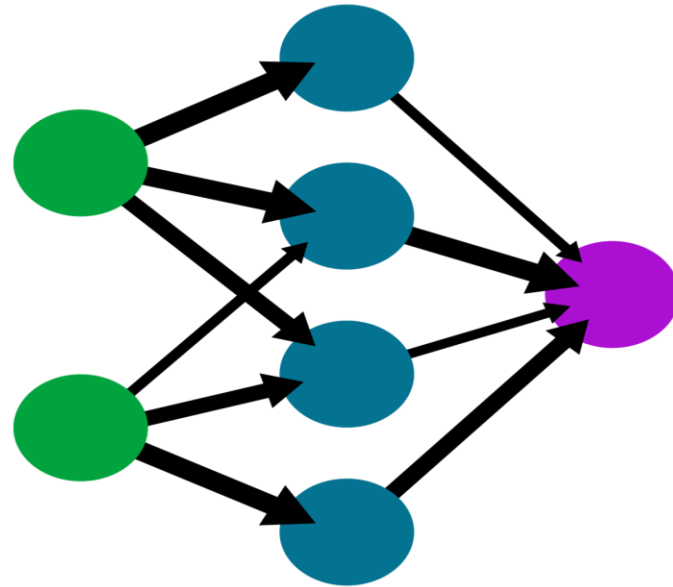
Neural Networks

A simple neural network

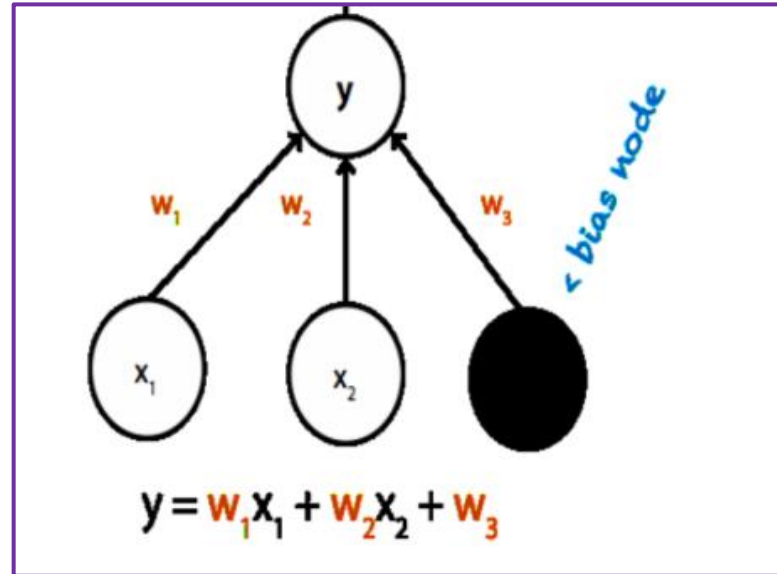
input
layer

hidden
layer

output
layer



Neural Networks



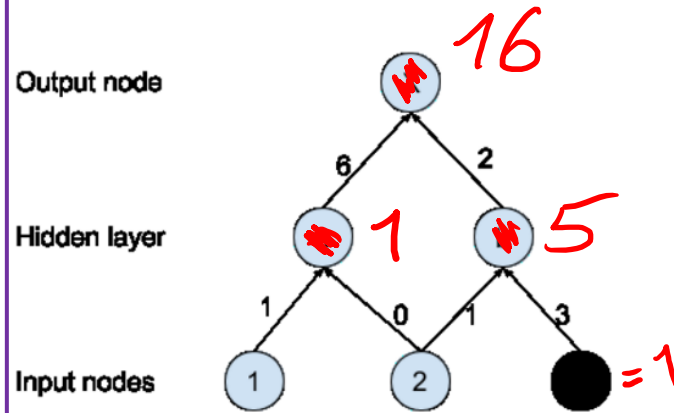
Bias node = 1

How to calculate the value of the output node?

`sum(values of children * their edge weight)`

Recap Question: Neural Networks

Here, we see a basic multilayer feedforward neural network, with weights and values on the input nodes. It has no activation function (all nodes are *linear*). The black node is a *bias node*. What is the value on the output node (marked X)?



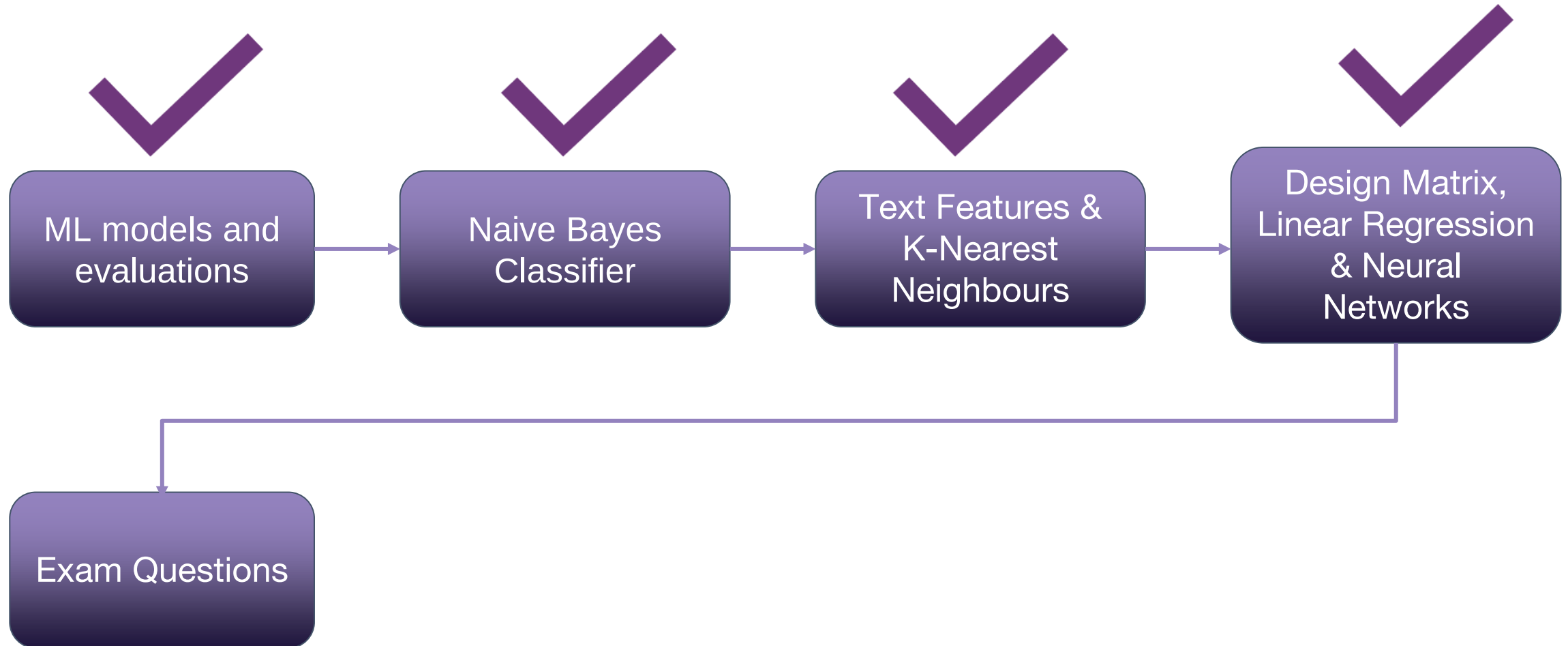
$$a = 1 \cdot 1 + 2 \cdot 0 = 1 + 0 = 1$$

$$b = 2 \cdot 1 + 3 \cdot 1 = 5$$

$$X = 6 \cdot 1 + 2 \cdot 5 = 16$$

$$X = 16$$

Planning Day 2



Final preparation: Exam Questions

Exam question: Probability

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

Given the full joint distribution as above, calculate the following:

$P(\text{Cavity} \mid \text{toothache} \vee \text{catch})$

$$P(\text{cavity} \mid + \vee c) = (0.108 + 0.012 + 0.072) / 0.416 = 0.46$$

$$P(\neg \text{cavity} \mid + \vee c) = (0.016 + 0.064 + 0.144) / 0.416 = 0.54$$

$$\Rightarrow P(+ \vee c) = 0.416$$

$$\langle 0.46, 0.54 \rangle$$

Exam question: Probability

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

$P(\text{Cavity} \mid \text{toothache} \vee \text{catch})$

$$P(\text{toothache}, \text{catch}) = 0.416$$

$$P(\text{cavity} \mid t, c) = (.108 + .012 + .072) / 0.416 = 0.4615$$

$$P(\neg \text{cavity} \mid t, c) = (.016 + .064 + .144) / 0.416 = 0.5385$$

$$P(\text{Cavity} \mid \text{toothache}, \text{catch}) = \langle 0.4615, 0.5385 \rangle$$

Exam question: Probability

Which of the following expressions is **not symmetric**?

A) Union probability: $P(A \cup B)$ is not always equal to $P(B \cup A)$

B) Joint probability: $P(A, B)$ is not always equal to $P(B, A)$

☒ C) Conditional probability: $P(A|B)$ is not always equal to $P(B|A)$

Exam question: Probability

Which of the following expressions is **not symmetric**?



A) Union probability: $P(A \cup B)$ is not always equal to $P(B \cup A)$ ✓

B) Joint probability: $P(A, B)$ is not always equal to $P(B, A)$ ✓

$$\begin{array}{r} 0.5 + 0.2 \\ 0.2 + 0.5 \\ \hline 0.7 \end{array}$$

☒ C) Conditional probability: $P(A|B)$ is not always equal to $P(B|A)$ ✗

Exam question: Joint Probability Distribution

Question 7 – Multiple response – Question-ID: 217964 (3 points)

Consider the following full joint probability distribution:

P	Q	R	P(P, Q, R)
true	true	true	0.26
true	true	false	0.33
true	false	true	0.04
true	false	false	0.14
false	true	true	0.03
false	true	false	0.08
false	false	true	0.11
false	false	false	0.01

Which statements are true? (2 right)

A $P(P = \text{true} \vee Q = \text{true} \vee R = \text{true}) = 1$

B $P(R = \text{false}) = 0.56$

C $P(Q = \text{true} | R = \text{true}) = P(R = \text{true} | Q = \text{true})$

D $P(Q = \text{true}) = 0.26$

E $P(P = \text{true} \ \& \ Q = \text{true}) = 0.59$

F $P(R = \text{false}) = 0.63$

Exam question: Joint Probability Distribution

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false	true	false	0.08
false	false	true	0.11
false	false	false	0.01

= 0.59

Which statements are true?

- ☒ A $P(P = \text{true} \vee Q = \text{true} \vee R = \text{true}) = 1$ $\rightarrow 0.99$
- ☒ B $P(R = \text{false}) = 0.56$
- ☒ C $P(Q = \text{true} | R = \text{true}) = P(R = \text{true} | Q = \text{true})$
- ☒ D $P(Q = \text{true}) = 0.26$
- ☒ E $P(P = \text{true} \ \& \ Q = \text{true}) = 0.59$
- ☒ F $P(R = \text{false}) = 0.63$

Exam question: Bayes' Rule

You are planning a picnic today, but the morning is cloudy

- 50% of all rainy days start off cloudy. $P(C|R) = 0.5$
- But cloudy mornings are common, as about 40% of days start cloudy. $P(C) = 0.4$
- This month is usually a dry month (only 10% of the days tend to be rainy) $P(R) = 0.1$

What is the chance of rain during the day given the morning is cloudy?

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$P(R|C) = \frac{0.5 \cdot 0.1}{0.4} = 0.125$$

Exam question: Bayes' Rule

You are planning a picnic today, but the morning is cloudy

- 50% of all rainy days start off cloudy.
- But cloudy mornings are common, as about 40% of days start cloudy.
- This month is usually a dry month (only 10% of the days tend to be rainy)

What is the chance of rain during the day given the morning is cloudy?

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\begin{aligned} P(\text{rain}|\text{cloudy}) &= P(\text{cloudy}|\text{rain}) * P(\text{rain}) / P(\text{cloudy}) \\ &= 0.5 * 0.1 / 0.4 \\ &= 0.125 \end{aligned}$$

Exam question: Precision & Recall

Given the following confusion matrix:

	actual +	actual -
predicted +	240	60
predicted -	80	170

What is the precision for this classification?

What is the recall for this this classification?

Exam question: Precision & Recall

Given the following confusion matrix:

actual + actual -

predicted + 240 60

predicted - 80 170

What is the precision for this classification?

0.8

What is the recall for this this classification?

0.75

Evaluation form



Step 1

Scan the QR code
or check your email



Step 2

Answer the 5
questions



Step 3

Sit back & relax



Exam question: ML Models

We would like to create a very simple self-driving car: we mount a low-resolution camera in front of the windscreen and simple sensors in the steering wheel. This way we can observe the way a human driver reacts to the road in front of him. We gather some data and set up a machine learning system. Which approach makes sense?

- ☐ We model this as a classification problem. The instances are the individual frames of the videofeed, the features are the pixel values of each frame, and the classes are 'left', 'right' and 'straight', depending what the driver was doing at the time the frame was recorded.
- ☐ We model this as a clustering problem. The instances are the individual frames of the videofeed, the features are the pixel values of each frame and the target value is the speed of the car.
- ☐ We model this as a clustering problem. The instances are the other cars on the road. The features are their speed, color, license plates and number of occupants. There is no target value, because clustering is an unsupervised learning method.
- ☐ We model this as a regression problem. The instances are the other cars on the road. The features are the driver's actions at a given moment in time, and the target values are the pixels in each frame.

Exam question: ML Models

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- ☐ We model this as a regression problem. The instances are the other cars on the road. The features are the driver's actions at a given moment in time, and the target values are the pixels in each frame.

Exam question: Error value

Consider a regression function $f(x) = x$ that was trained on some training set, and that is now evaluated with the following test-set of datapoints $\{ (2,2), (4,3), (4,3.5), (4,5), (1,1), (3,2) \}$.

What is the value of the standard squared error function.

Exam question: Text Features

Consider the sentence:

“In the true Computer Sciences, where science rules, a rule computes truth.”

Which operations have been applied to this sentence when it has been reduced to the following set of features:

< tru, comput, scien, rule>

- ☐ Stemming
- ☐ Summarization
- ☐ Remove capital letters
- ☐ Remove stop words
- ☐ Remove punctuation
- ☐ Remove part of speech
- ☐ Sentiment analysis
- ☐ Parsing
- ☐ Noun tagging

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- ☐ Sentiment analysis
- ☐ Parsing
- ☐ Noun tagging

Exam question: Feature Vectors

Given three feature vectors

1. $\langle 1, 1, 1, 1, 1, 0, 0, 1, 0, 0 \rangle$
2. $\langle 0, 1, 1, 0, 0, 1, 1, 1, 0, 0 \rangle$
3. $\langle 0, 1, 0, 1, 1, 0, 0, 0, 1, 1 \rangle$

Which vector represents the nearest neighbour of the vector

$\langle 0, 1, 0, 0, 1, 1, 0, 1, 0, 0 \rangle$

if we apply Euclidean distance?

- A. 1
- B. 3
- C. none
- D. 2

Exam question: Feature Vectors

Given three feature vectors

1. $\langle 1, 1, 1, 1, 1, 0, 0, 1, 0, 0 \rangle$
2. $\langle 0, 1, 1, 0, 0, 1, 1, 1, 0, 0 \rangle$
3. $\langle 0, 1, 0, 1, 1, 0, 0, 0, 1, 1 \rangle$

Which vector represents the nearest neighbour of the vector

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if we apply Euclidean distance?

- A. 1
- B. 3
- C. none
- ☒ D. 2

Exam question: Linear Regression

Let us try to fit a linear function to a number of training sets where the slope s is known to be -1 . Suppose we have three pairs of values $(0,0)$, $(1,1)$ and $(2,-1)$ for two features. What is the optimal fitting function in this model class?

- A. $f(x) = -x + 1$
- B. $f(x) = -x$
- C. $f(x) = -x + 2$

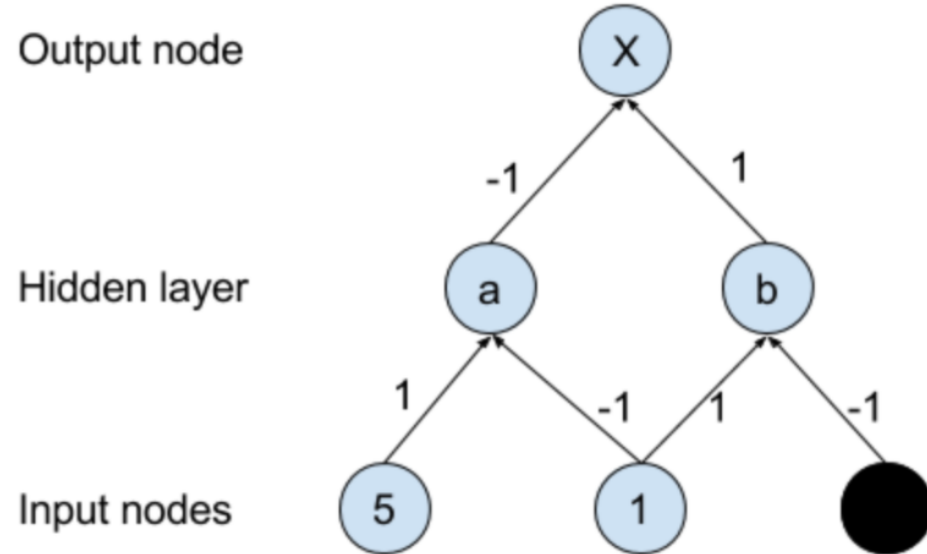
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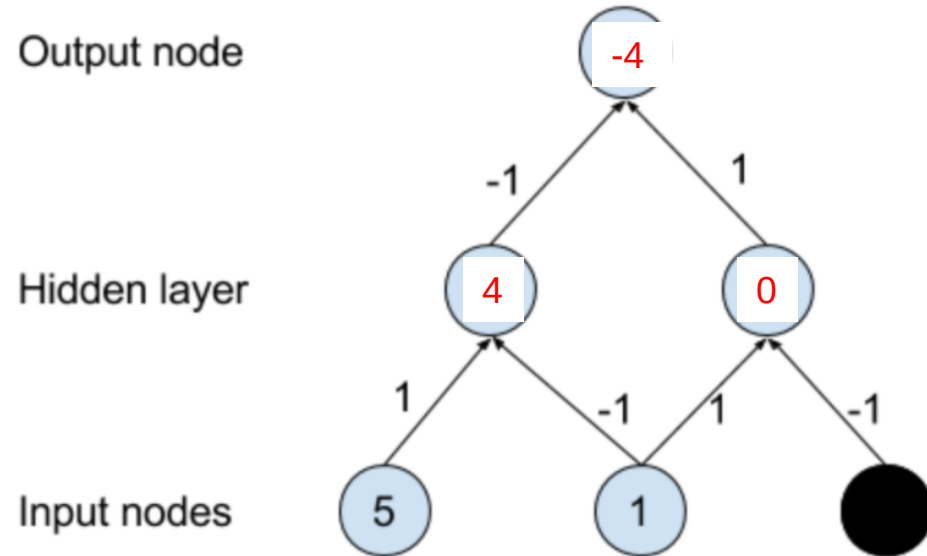
Exam question: Neural Networks

Here, we see a basic multilayer feedforward neural network, with weights and values on the input nodes. It has no activation function (all nodes are *linear*). The black node is a *bias node*. What is the value on the output node (marked X)?



Exam question: Neural Networks

Here, we see a basic multilayer feedforward neural network, with weights and values on the input nodes. It has no activation function (all nodes are *linear*). The black node is a *bias node*. What is the value on the output node (marked X)?



Exam question: Naive Bayes

(6 Points) Consider the following dataset:

word1	word2	class
T	T	spam
T	F	spam
T	T	ham
T	T	ham
F	T	spam
T	T	ham
T	T	ham
F	T	spam

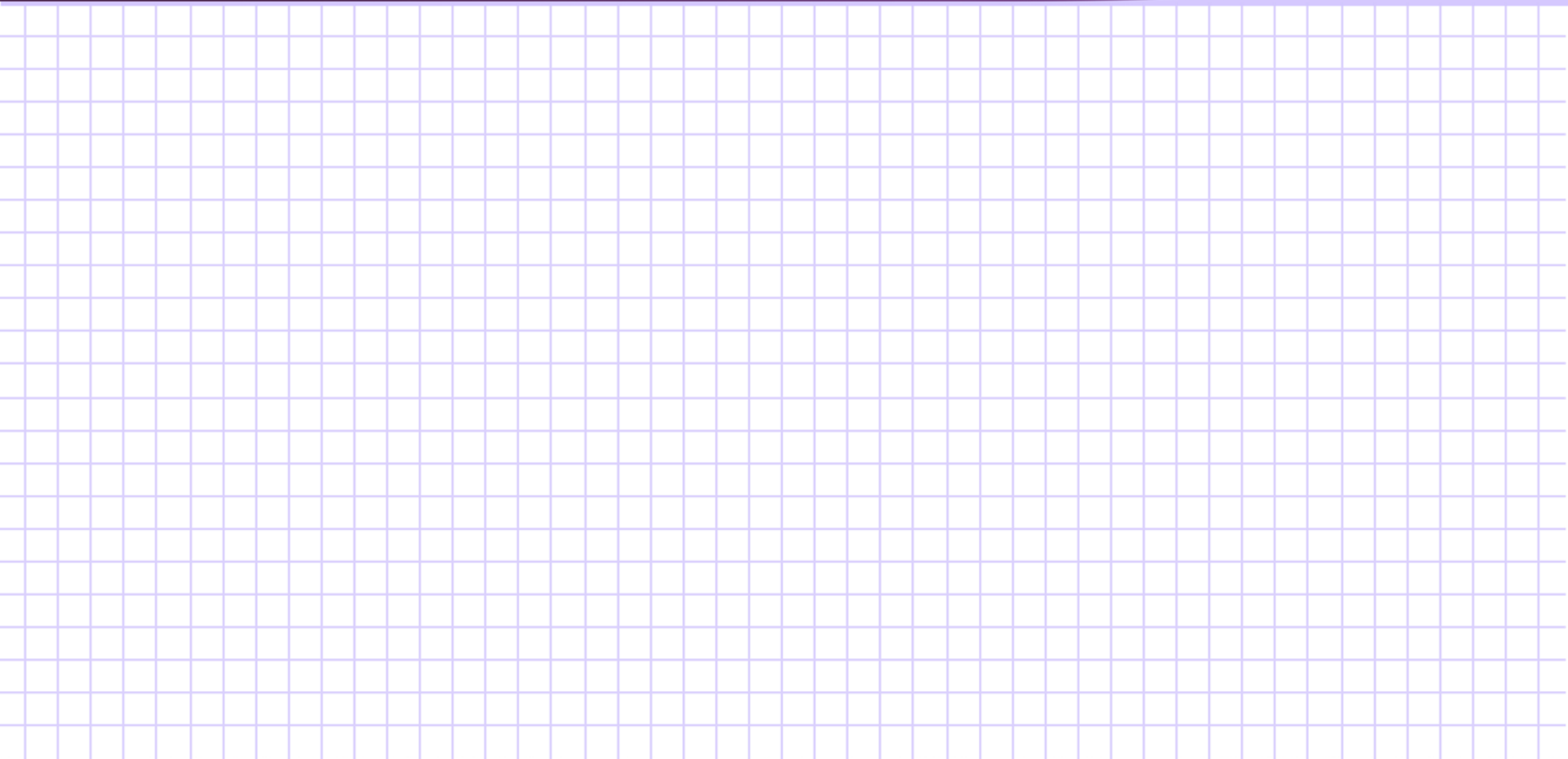
Each row represents an e-mail. The feature `word1` indicates for some word whether or not it occurs in the email (T) or not (F). The same with feature `word2`. These features are the only ones recorded in the data. We will be using a **naive bayes classifier** to classify a new email containing `word1`, but not `word2`.

We estimate $p(\text{spam})$ and $p(\text{ham})$ from the dataset.

Which of the following statements is true?

- A. $P(\text{spam} | \text{word1}, \neg \text{word2}) > P(\text{ham} | \text{word1}, \neg \text{word2})$
- B. $P(\text{spam} | \text{word1}, \neg \text{word2}) < P(\text{ham} | \text{word1}, \neg \text{word2})$
- C. $P(\text{spam} | \text{word1}, \neg \text{word2}) = P(\text{ham} | \text{word1}, \neg \text{word2})$

Exam question: Naive Bayes



Exam question: Naive Bayes

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word1	word2	class
T	T	spam
T	F	spam
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F	T	spam
T	T	ham
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F	T	spam

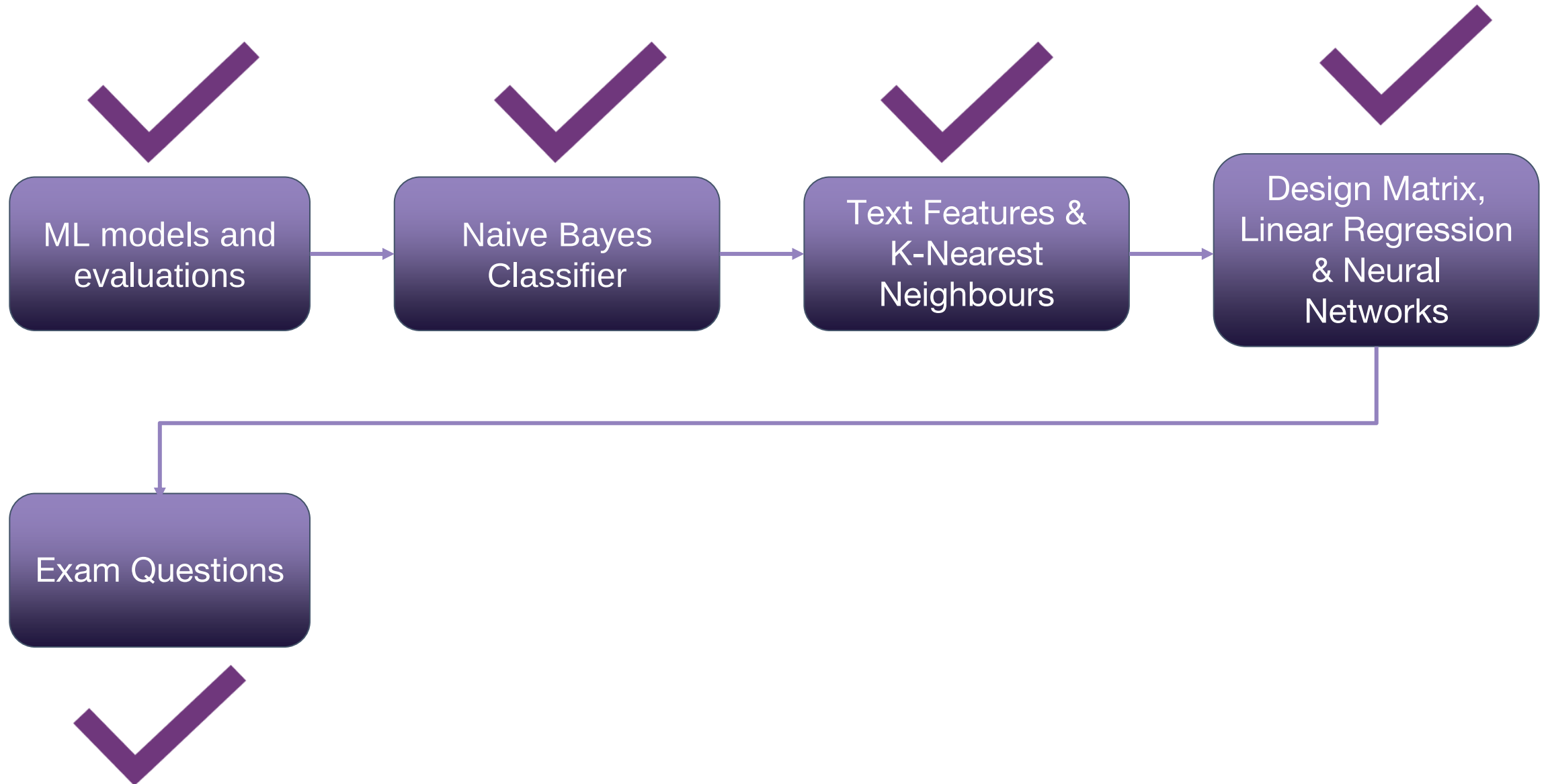
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Planning Day 2



Ethics: open question

- Asimov's Laws
- The trolley problem
- Autonomous weapons
- Privacy and AI
- Racism and AI
- Limits of AI

These are not all topics! Refer to the lecture slides for this!

Example Exam Questions: Ethics

(3 Points) Briefly, in 6 lines, explain possible ethical differences between the trolley dilemma (as discussed in the lecture) and the dilemma faced by an engineer building a decision module for an automatic car that has to decide between the life of its driver or a crowd it might run into.

(4 Points) Consider the statement: "There is no need to worry about the ethical implications of AI. It'll be at least decades before it becomes a problem, and we don't even know if it's possible yet."

Write a small essay arguing in favor of this statement or against it. You will not be graded on which position you take. Don't write down every thought you have on the subject, focus on making a single, clear argument.

You will be judged on how clear, how well written and how sound your argument is.

Example Exam Questions: Ethics

Consider the statement: "Strong artificial intelligence will never be achieved. There is an undefinable quality to the human mind that could never be captured by the simple rule-based operation of a computer program."

Write a small essay arguing in favor of this statement or against it. You will not be graded on which position you take. Don't write down every thought you have on the subject, focus on making a single, clear argument. You should aim for between 200 and 400 words

You will be judged on how clear, how well written and how sound your argument is.

Example Exam Questions: Ethics

Which of these are reasonable objections to the Turing test (multiple answers possible)

- A** it is 70 years old, so it no longer applies to modern AI
- B** it imposes a white male dominant view on the notion of intelligence
- ☒ **C** it uses human intelligence as the standard for intelligence
- D** it degrades humans by comparing them to machines

Example Exam Questions: Ethics

Which of the following problems are related to the various variants of the Trolley dilemma as presented in the lecture in the context of Intelligent Systems?

- A** A variant of the trolley problem is when someone has to programme a decision procedure for a self-driving car, which has to decide between crashing into a crowd of people, or killing the driver (hitting a wall, or falling down a cliff)?
- B** A variant of the trolley problem is when someone has to decide whether or not to buy a trolley ticket. AI research has shown that one can optimise the costs of travelling by not paying for a ticket, and instead paying the extra fee in case of a control.
- C** A variant of the trolley problem is when someone has to programme a decision procedure for a trolley to stay within the schedule of the tram-line, or to keep within the operational rules of the transportation system the trolley operates in.
- D** A variant of the trolley problem is when someone has to decide whether or not to enter a fully automated trolley, which is driven through a complex traffic environment by AI technology.
- E** A variant of the trolley problem is when someone has to programme a decision procedure for a fully automated trolley to decide on who to provide access to the trolley in case of restricted space.

Example Exam Questions: Ethics

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- A** it is 70 years old, so it no longer applies to modern AI
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Evaluation form



Step 1

Scan the QR code
or check your email



Step 2

Answer the 5
questions



Step 3

Sit back & relax



Thank you and good luck with your exam!

A handwritten message in black ink on a white background. The text "Good Luck!" is written in a cursive style. Below the text is a long horizontal line that slopes upwards from left to right. To the right of the line is a simple smiley face drawn with a circle and two dots for eyes and a curved line for a mouth.